

How to succeed

- Read the Chapter ahead of lectures
- Come to class
- **Start paper early**
- Study groups
- Prepare exams
- Don't cheat
- **Learn with Internet: Wikipedia, google(baidu)...**
- Ask questions and Think skeptically

Introduction to Polymer Materials

Lecture 9: Adhesives

SUN Dazhi

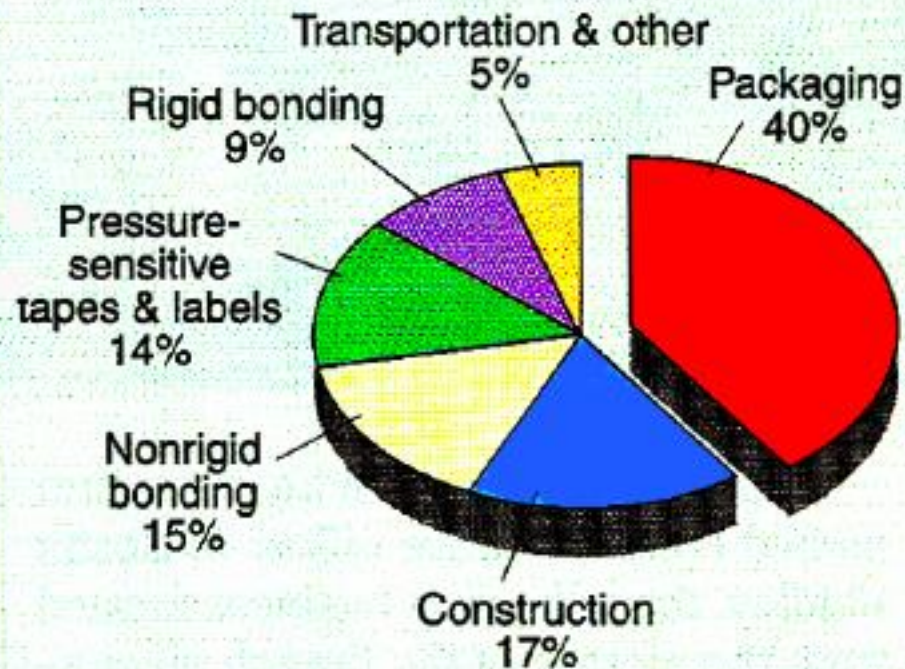
Adhesive

- A substance capable of holding materials together by **surface attachment**.
- A general term that includes cement, glue, mucilage and paste.



Markets for Adhesives

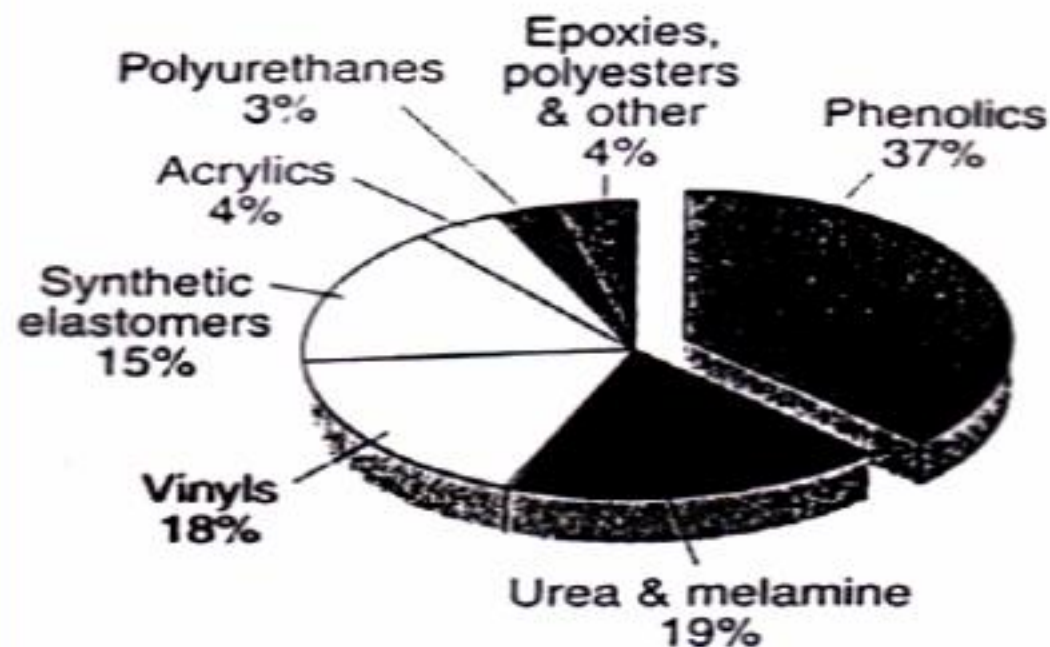
Packaging leads \$9 billion adhesives market . . .



1999 demand = 5.9 billion lb

Chemical Types of Adhesives

Largest group of synthetic adhesives are based on phenolic polymers



1996 U.S. demand = 8.47 billion lb

Source: Freedonia Group

History of Adhesives

- The first adhesives were gums and other plant resins.
 - Vegetable gums:
 - ❖ Guar gum
 - ❖ Gum Arabic
- Archaeologists have found 6000-year-old ceramic vessels that had broken and been repaired using plant resin.
- Most early adhesives were animal glues made by rendering animal products.
 - Native Americans use of buffalo hooves
 - Hide glue
 - Bone glue
 - Fish glue
 - Rabbit skin glue

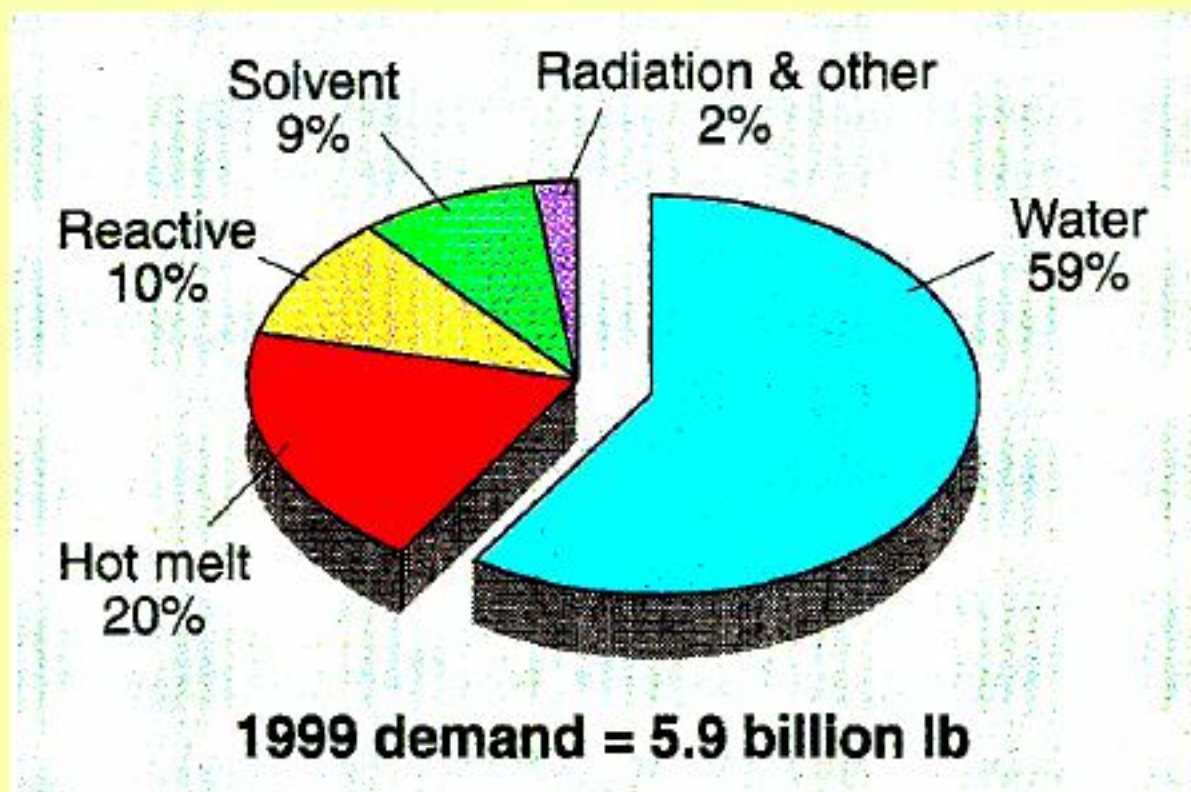
History--cont.

- Native Americans in what is now the eastern United States used a mixture of spruce gum and fat as adhesives and as caulk to waterproof seams in their birch bark canoes.
- During the times of Babylonia, tar-like glue was used for gluing statues.
- Egypt was one of the most prominent users of adhesives.
 - The Egyptians used animal glues to adhere tombs, furniture, ivory, and papyrus.
- Mongols used adhesives to make their short bows.
- In Europe in the Middle Ages, egg whites were used to decorate parchments with gold leaves.
- In the 1700s, the first glue factory was founded in Holland, which manufactured hide glue.
- In the 1750s, the British introduced fish glue.
- As the modernization continued, new patents were issued by using rubber, bones, starch, fish, and casein.

Types of Adhesives

- **Solvent based adhesives**
 - Contact adhesives
 - Rubber cement
- **Reactive**
 - Thermosets (epoxy, uv cured, water cured)
 - Anerobic
 - superglue
- **Hot melts**

Application (Liquid) Methodology



Solvent Based or Drying Adhesives

- White glue
 - Latex of polyvinyl acetate
- Contact adhesives (e.g. liquid Nails)
 - Applied to both surfaces
 - Polyisoprene and polychloroprene in solvent
 - Or water based acrylics
- Rubber cement
 - Elastomer in organic solvent

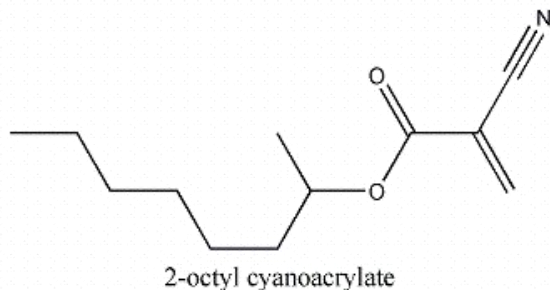
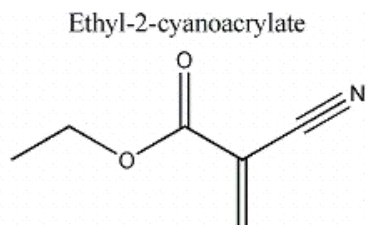
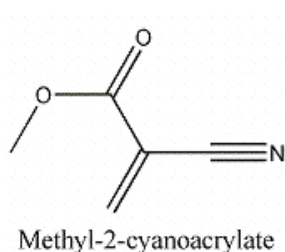


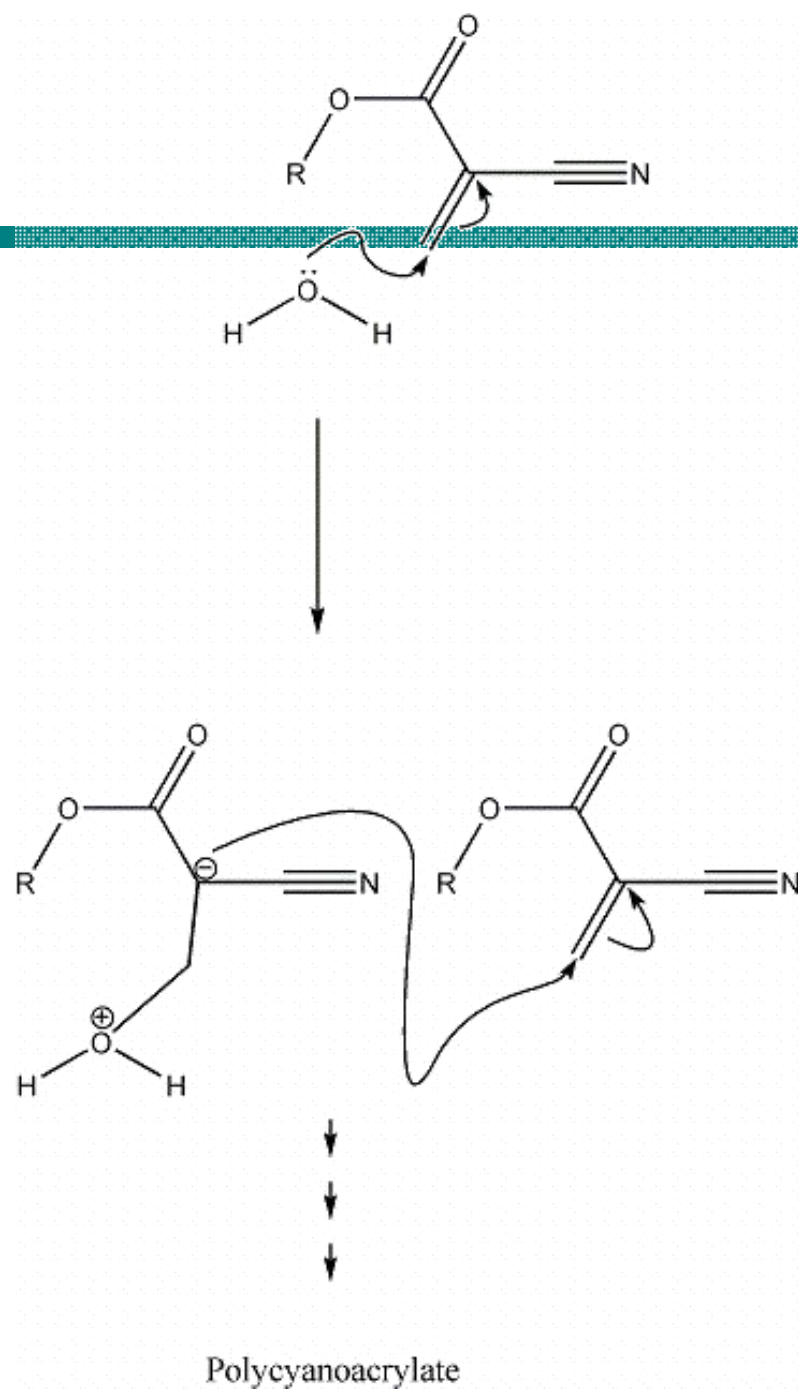
Reactive adhesives

- Cyanoacrylates-superglues
- Thermosets
- Anerobics
- UV cured

Superglue

- cyanoacrylates - polycyanoacrylates
- Polymerize with water or other nucleophiles (amines, thiols, etc.) to generate bond
- Water on every surface
- Thermoplastic product (soluble in acetone)





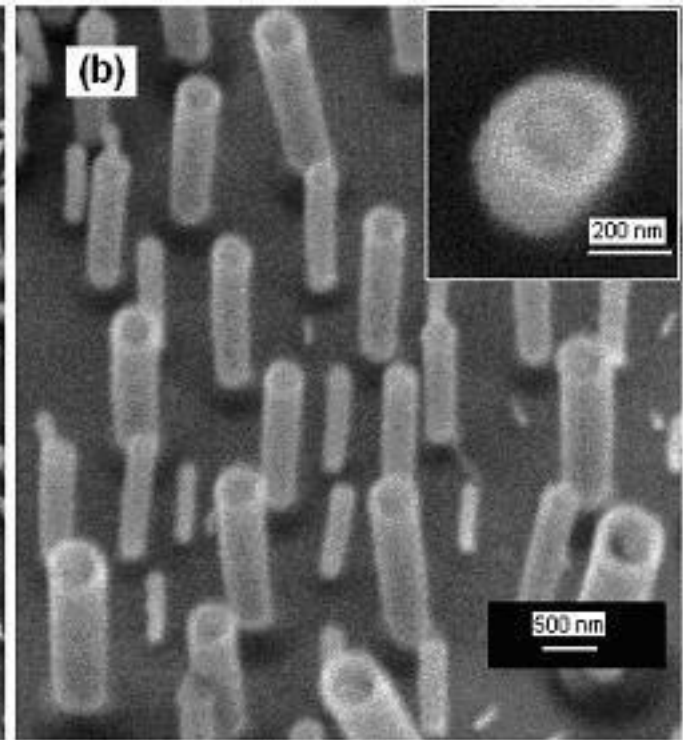
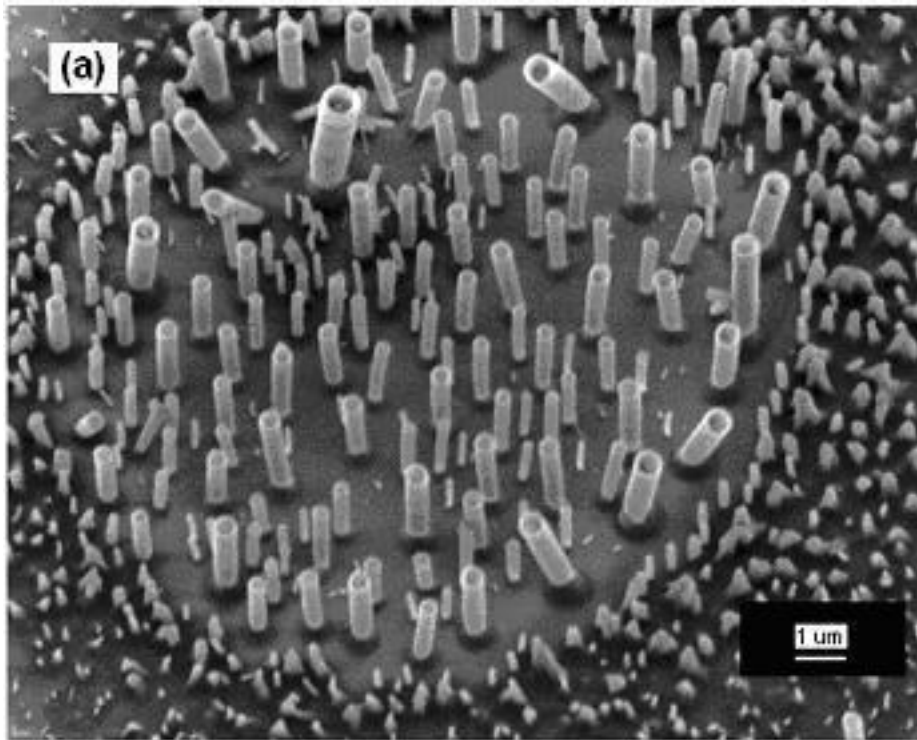
Superglue for biological applications



Superglue and Fingerprints



Superglue Nanofibers (close-up of fingerprints)

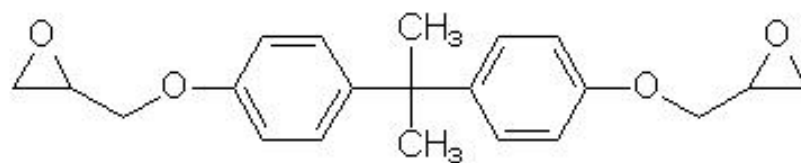
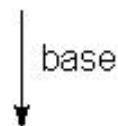
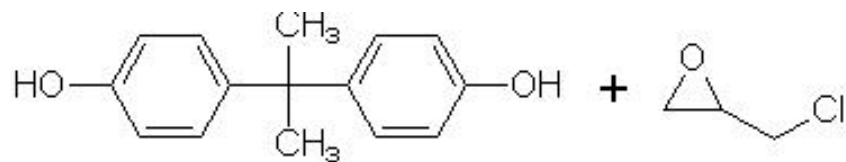


Thermoset Adhesives

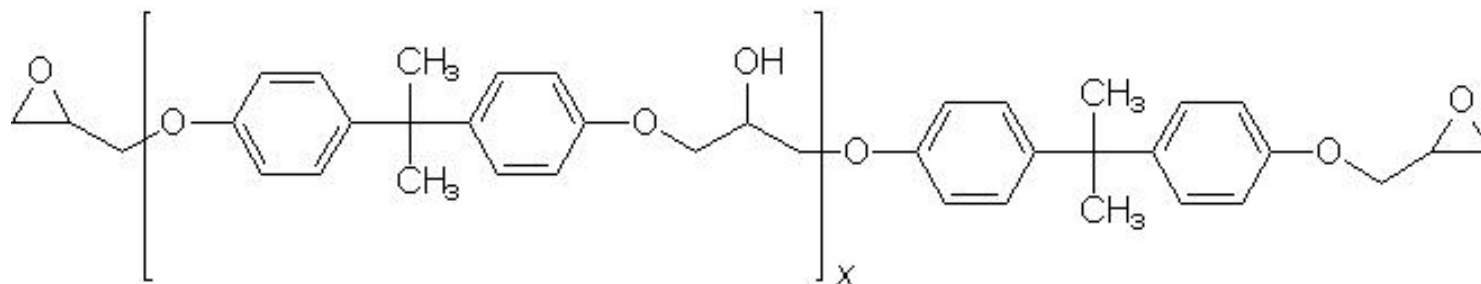
- Low molecular weight precursors (liquid)
- Good wetting of surfaces
- React to form highly crosslinked glassy material
- Surface interactions & mechanical interlocking
- Can be highly filled
- Phase separation for toughness
- Epoxies, polyesters, urethanes (PUR), even silicones

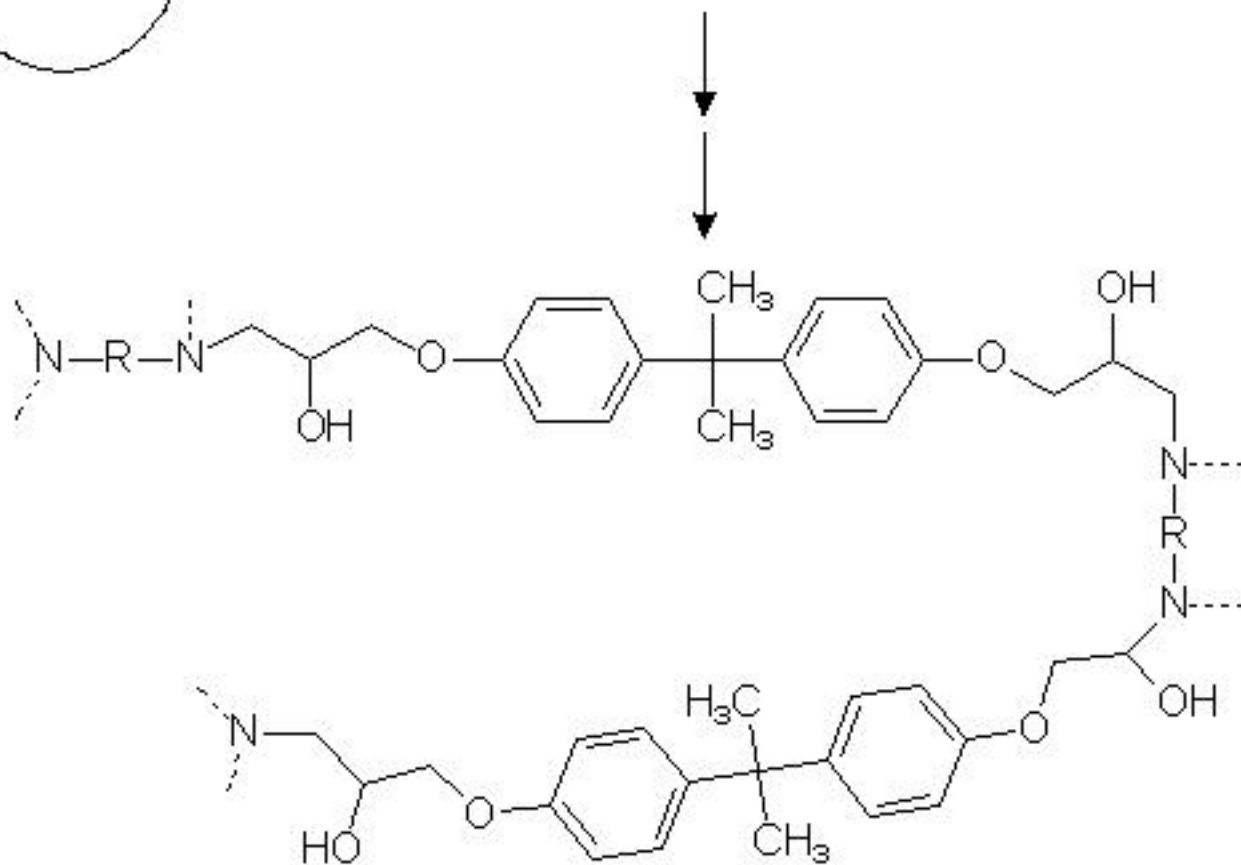
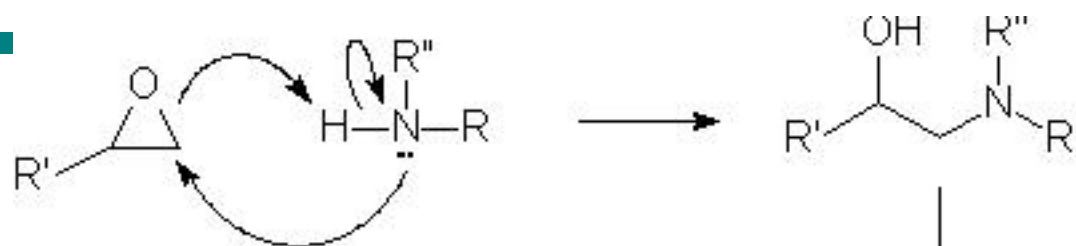


Epoxy Adhesives



or





Epoxy



Special epoxy is strong enough to withstand the extreme force transferred from a surfboard fin to the fin mount. This epoxy is waterproof and capable of curing underwater. The blue coloured epoxy on the left is still undergoing curing.



Terrazzo with adapted Native-American design at the Hoover Dam



An epoxy encapsulated hybrid circuit on a printed circuit board.

A Case Study

Epoxy Failure in Boston Tunnels



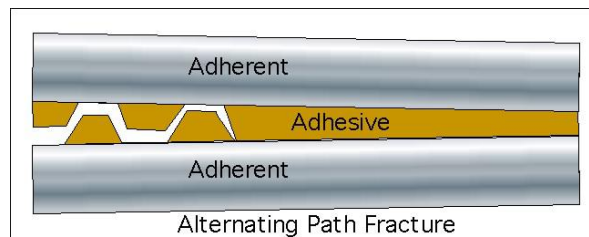
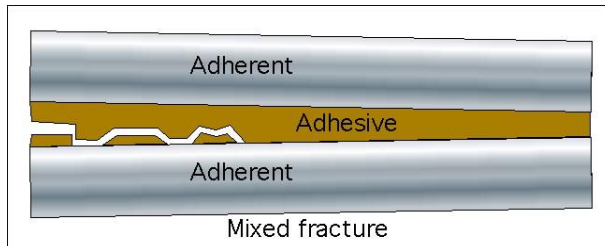
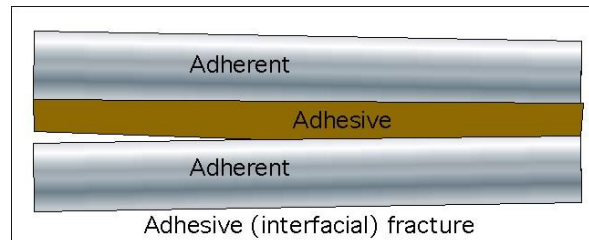
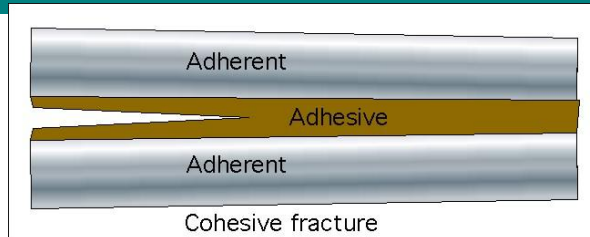
BIG DIG BOLTS

How they work, how they failed



On July 10, 2006, concrete ceiling tiles in the Interstate 90 connector tunnel under South Boston fell on a passing car, killing Milena Del Valle and injuring her husband, Angel. Investigators are focusing on a failure of the bolts attaching the panels to the tunnel roof as a possible cause of the accident. [Learn about the system that failed >>](#)

Adhesive Failure



- Adhesives can fail at several different points.
- Common failures are:
 - Cohesive
 - Adhesive (Interfacial)
 - Mixed fracture
 - Alternating crack path

MATERIAL AND INSTALLATION SPECIFICATIONS

Physical Properties for Adhesive

Shelf Life	2 years for components
Storage Conditions	Store dry at 40° to 90°F
Injection Temperature	For best results, condition material to 60°F - 85°F
Color	Component A – White Component B – Dark Gray
Mixing Ratio	1:1 by volume
Mixed Consistency	Smooth, non-sag, uniform gray paste
Shore Hardness (ASTM D2240)	D 86 – 90
Compressive Strength (ASTM D 695)	11,125 psi, 1 day 14,740 psi, 7 days
Tensile Strength (ASTM D 638)	7,250 psi, Fast Set 7,400 psi, Standard Set
Flexural Strength (ASTM D 790)	6,200 psi, Fast Set 6,700 psi, Standard Set
Slant Shear Strength (ASTM D732)	4,900 psi, Fast Set – 1 day 6,700 psi, Standard Set – 14 day
Water Absorption (ASTM D 570)	Less than 1% (0.59%)
Bond Strength (Dry cure)	3,000 psi, concrete to concrete 3,000 psi, steel to concrete
Shrinkage (ASTM D 2566)	Not measurable
Heat Deflection (ASTM D 648)	140°F (60°C)

Setting Times

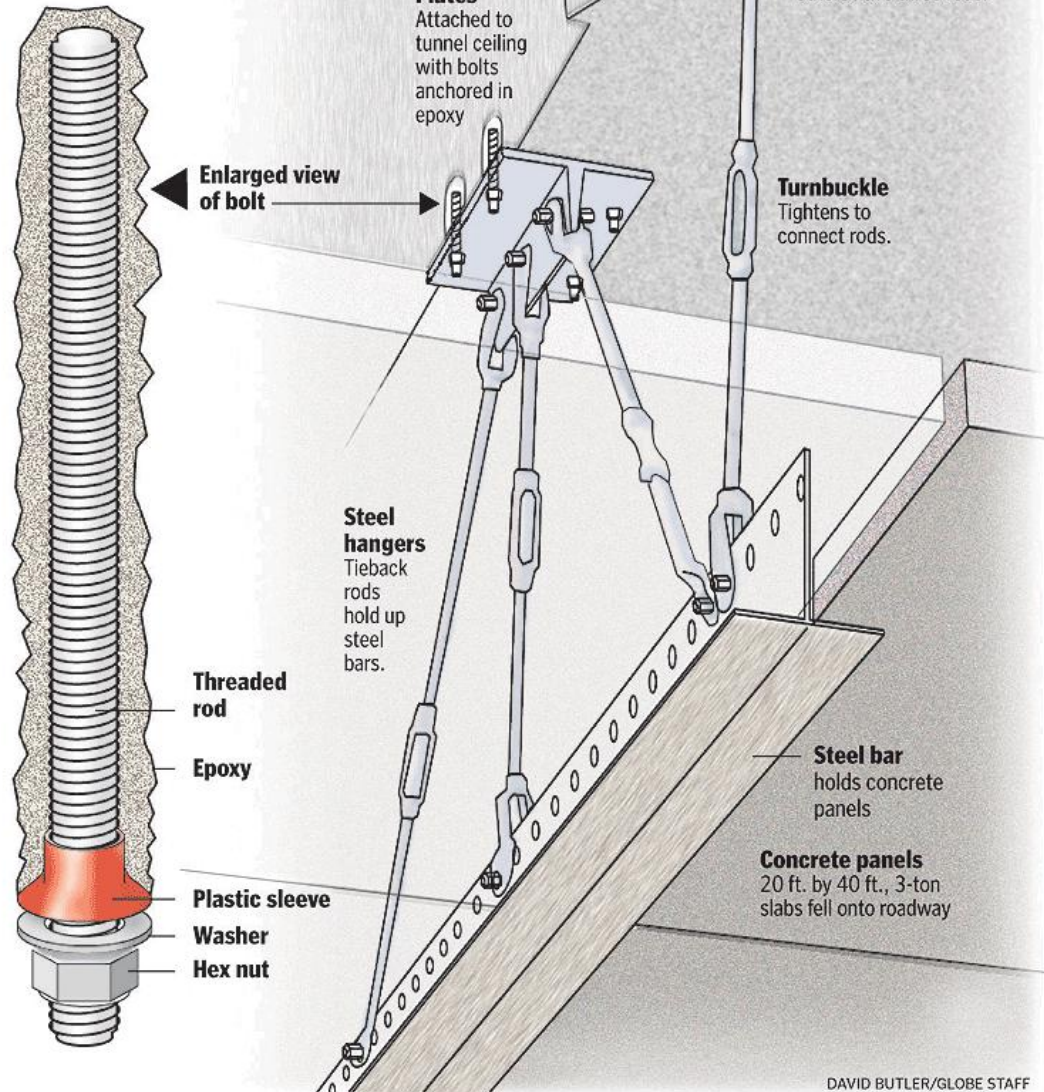
Base Material Temp. (°F)	Maximum Gel Time ^{1,2} (Minutes)		Minimum Curing Time ^{2,3} (Hours)		Full Curing Time ⁴ (Hours)	
	Fast Set	Std. Set	Fast Set	Std. Set	Fast Set	Std. Set
40	30	60	8	16	36	48
60	20	45	3	7	24	36
75	15	35	2	6	24	24
90	10	20	1 1/2	4	16	24
°C = 5/9 (°F-32)						

1. The gel time is the maximum time during which the epoxy can be worked before it begins to harden.
2. Anchors must not be disturbed between the maximum gel time and the minimum curing time.
3. The Power-Fast+ adhesive can support the weight of the anchor element (not including any fixture) at half of the minimum cure time.
4. The full curing time is the time required for the epoxy to harden and achieve published load capacities. Anchors may not be tightened until the full curing time has elapsed.



Installed bolts lengths were actually shorter than specifications

Investigators are looking at how concrete panels were fastened to the tunnel ceiling, including the quality of the bolts. A 1998 state report found problems with the glue and bolts on a nearby section of the Big Dig.



BIG DIG BOLTS

How they work, how they failed

THE OLD SYSTEM EPOXY-BASED FIXTURES

The old system

HOW THE EPOXY PROCESS SHOULD WORK

The glue used in the I-90 connector tunnel was a two-part adhesive that should harden into a tight chemical bond after components are mixed.



DRILLING

Holes were drilled in some cases with diamond-tip drill bits to cut through the reinforcement bars embedded in concrete. The epoxy manufacturer recommends carbide bits that leave a rougher hole, and a better bond for epoxy.

Holes should be cleaned before epoxy is applied.



EPOXY APPLICATION

Resin and hardener applied separately into drilled hole.

Drilling holes too deep or applying too little epoxy can cause failure.

SOURCE: 1998 State Inspector General report, engineers' field reports

◀ BACK NEXT ▶

BIG DIG BOLTS

How they work, how they failed

THE OLD SYSTEM EPOXY-BASED FIXTURES

The old system

HOW THE EPOXY PROCESS SHOULD WORK



MIXING

Screwing anchor bolt into hole mixes components.

Bolts must be clean for epoxy to adhere.

Mixed



HARDENING

Epoxy must be given enough time to cure.

Cold weather lengthens curing time.

Hardened



◀ BACK NEXT ▶

BIG DIG BOLTS

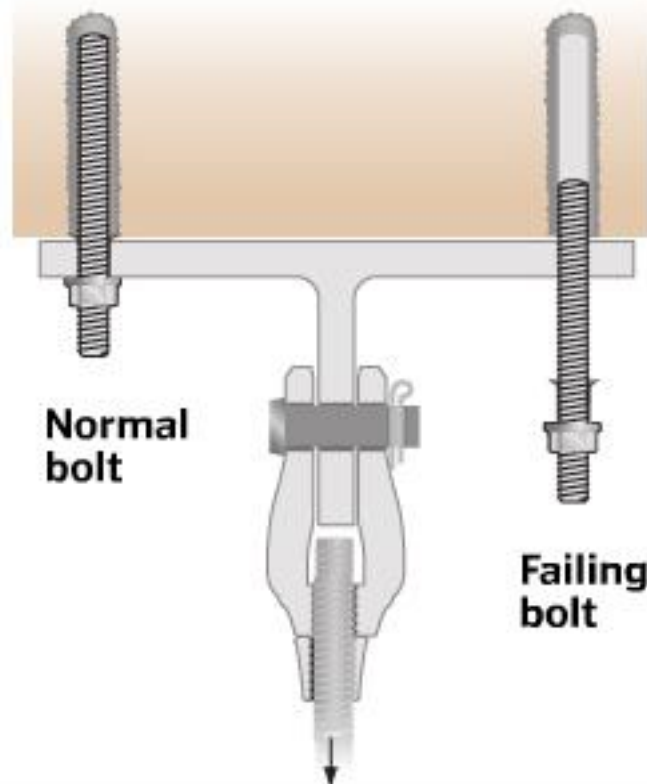
How they work, how they failed

THE OLD SYSTEM EPOXY-BASED FIXTURES

The old system

HOW THE BOLTS SEPARATED

Some of the failed bolts holding the concrete panels that fell pulled out cleanly, devoid of epoxy or concrete chunks. Further inspection has revealed some failing bolts with epoxy on them.



A bolt in the eastbound Ted Williams Tunnel on July 20.

◀ BACK TO START

◀ BACK

Figure 2 | Room temperature property trends vs. $T_g >$ test temperature for amine-cured epoxy resins.

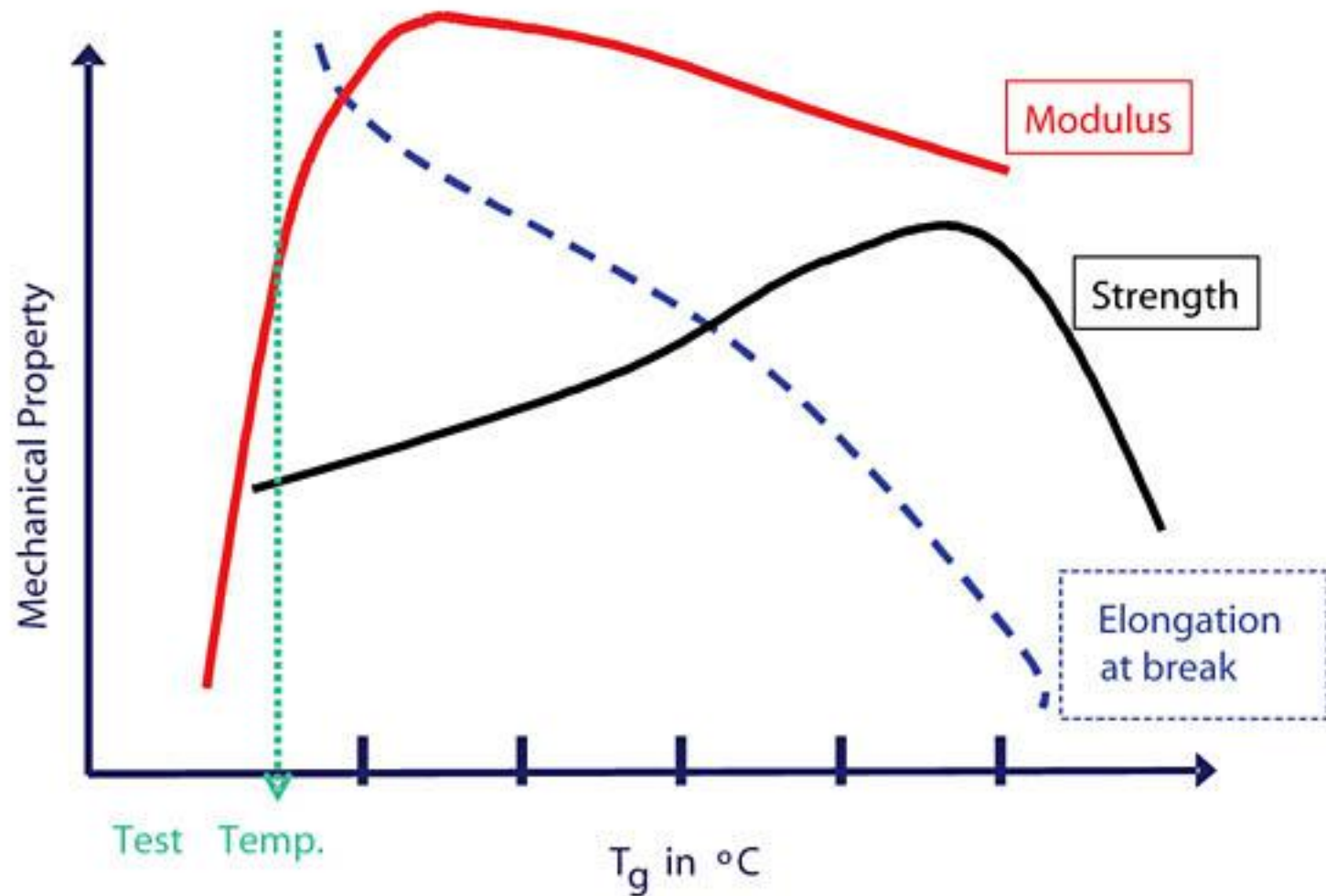


Table 1 | Relative T_g s of a DGEBA-type resin cured with various types of amines.

T_g Range (°C)	Amine Type	Examples of Amine Type
40-100	Polyetheramines (PEAs)	JEFFAMINE® T-403 amine JEFFAMINE® D-230 amine
40-100	Polyamides & Amidoamines	VERSAMID® 125 polyamide GENAMID® 490 amidoamine
110-125	Ethyleneamines	DETA (diethylenetriamine), TETA (triethylenetetramine), TEPA (tetraethylenepentamine), AEP (N-aminoethylpiperazine)
145-175	Cycloaliphatic	PACM (bis-(p-aminocyclohexyl) methane), DACH (diaminocyclohexane), DMCH (bis-(dimethyldi-aminocyclohexyl)methane)
160-220	Aromatic	MDA (methylene dianiline), m-PDA (m-phenylene diamine), DDS (diaminophenyl sulfone)

Slower curing resin: specified for Big Dig

Polyetheramines 25 – 100 degrees C Tg

Moderate Tg, thus flexible

Long pot-life

Slow curing, low exotherm

Only slight odor

Very low color

Low viscosity

Fast curing resin: used in Big Dig

Ethylene diamines and their Adducts

110 – 130 degrees C Tg

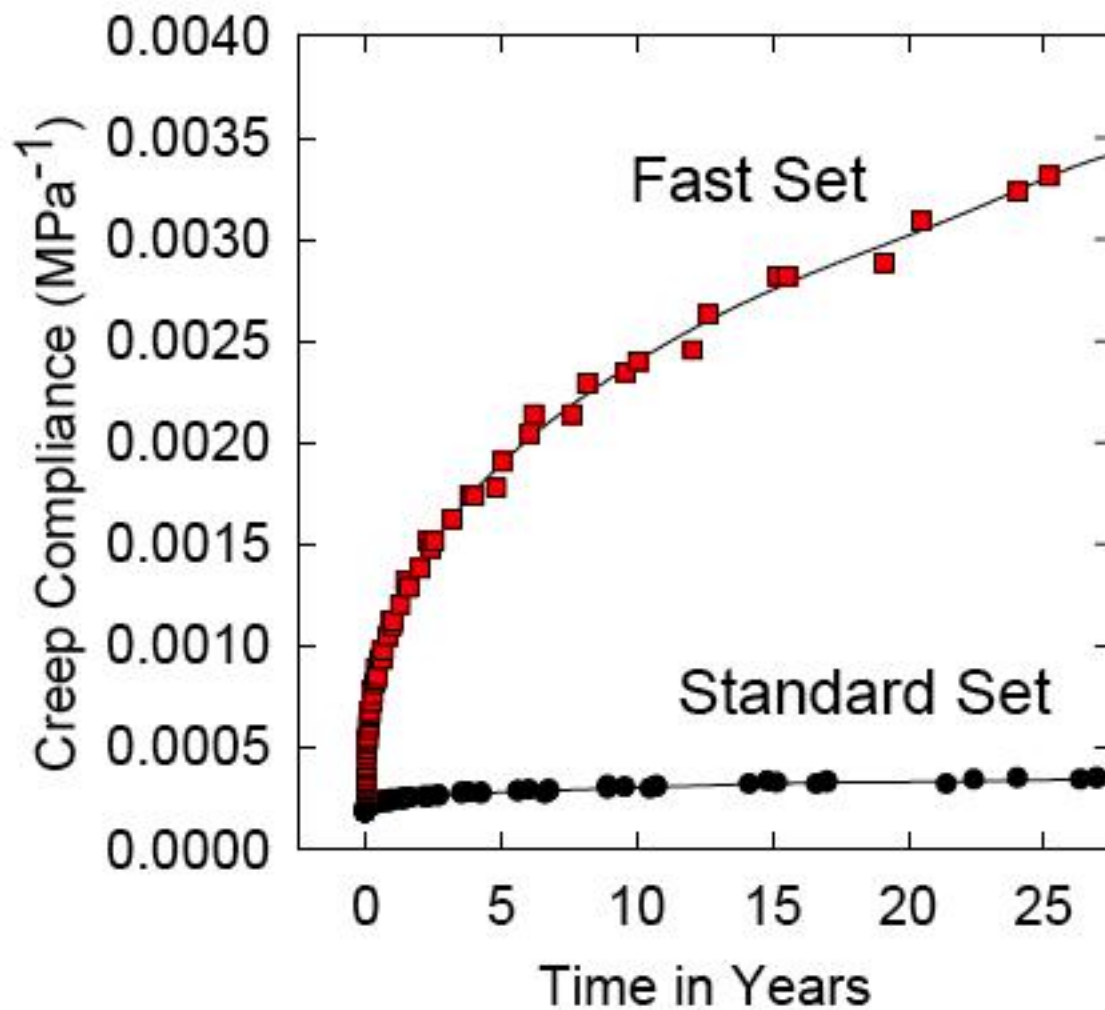
Somewhat brittle

Short pot-life

Fast curing, high exotherm

Handling concerns/odor

Low viscosity



↑
Increasing
anchor
displacement

Data courtesy
of NIST

Anchors loaded at 4000 pounds

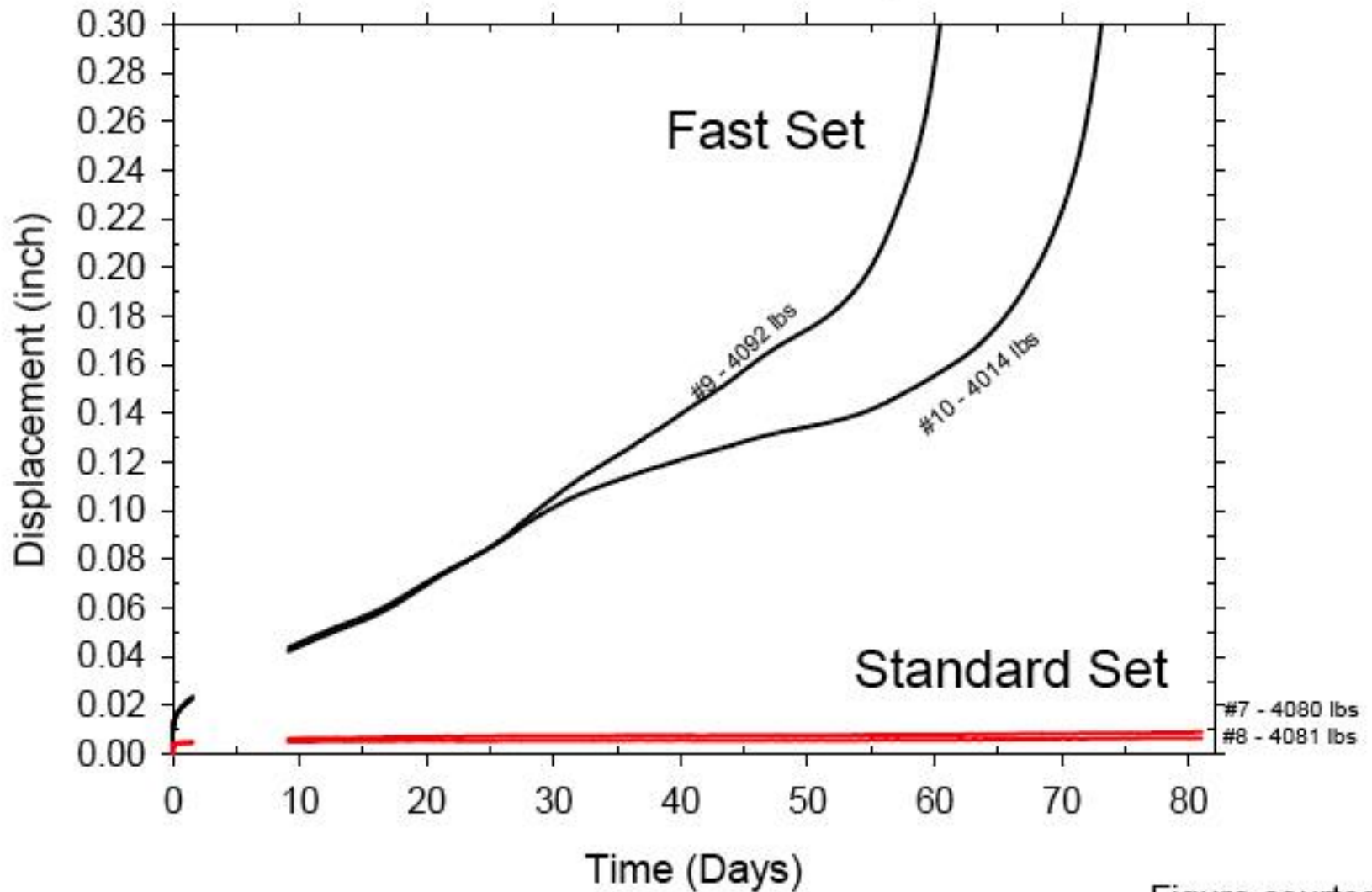
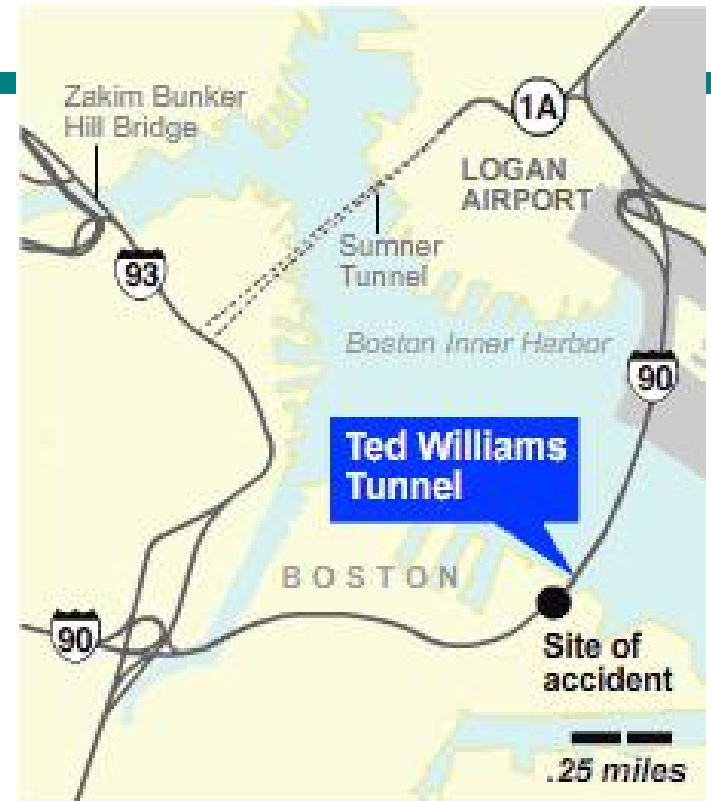


Figure courtesy
of FHWA TFHRC



Using the wrong epoxy led to long-term deformation and failure in the I-90 tunnel plenum that killed a 38-year-old local woman.

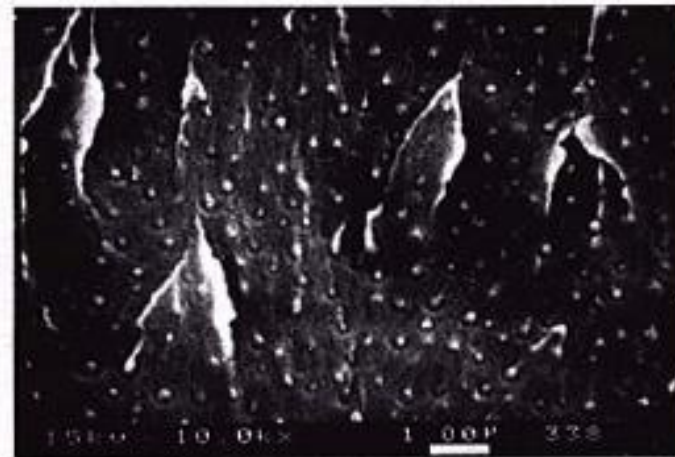
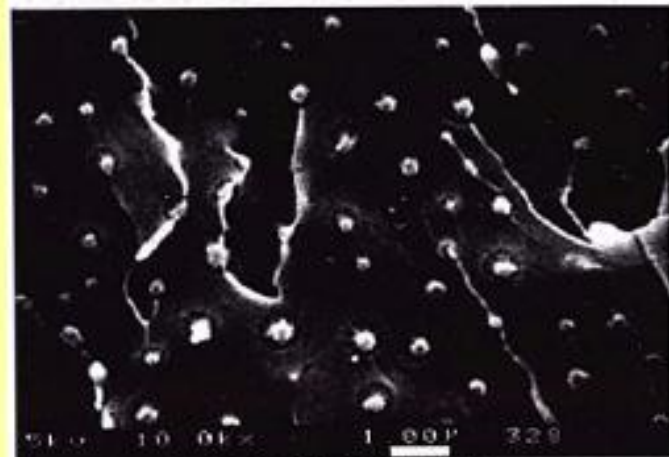
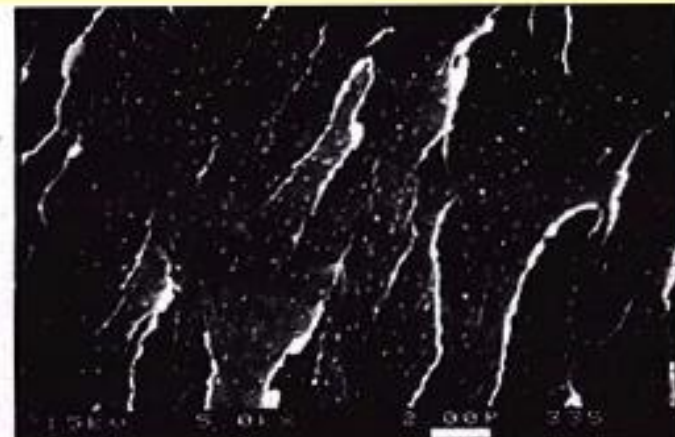
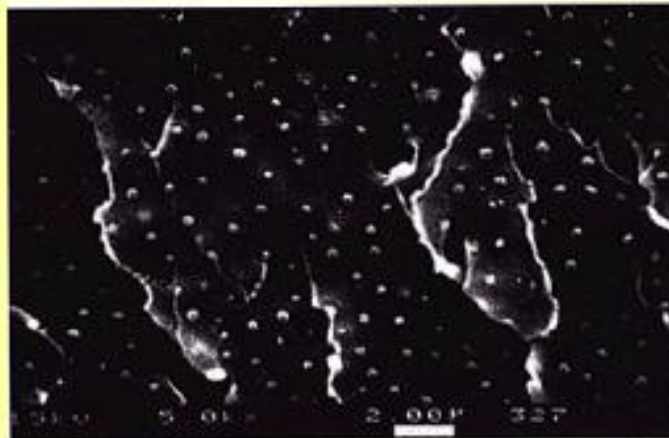


Poor creep resistance of fast set epoxy kills!!!

Strength of adhesives depends on cost

- Auto industry: used galvanized steel to increase rust resistance, but didn't have a good way to weld it. Tried to use adhesives to bond the galvanized steel.
- Bonds didn't last the 5-7 years target life, because they were using \$1/lb adhesives.
- Now automobiles use \$10/lb adhesives.
- Aircraft use \$400/lb adhesives.

Phase Sepn. in Epoxy-PSF Networks

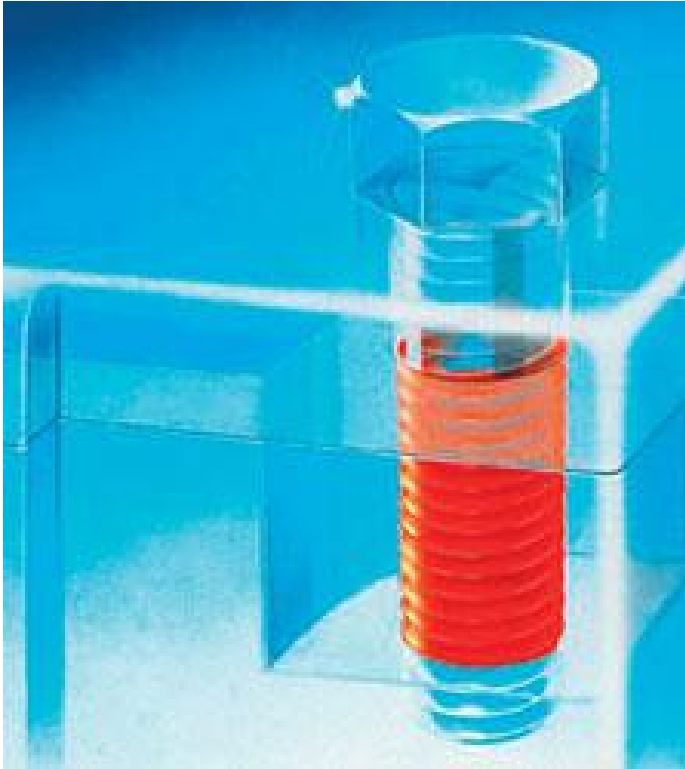


THERMAL CURED

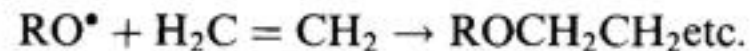
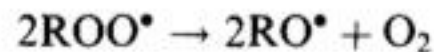
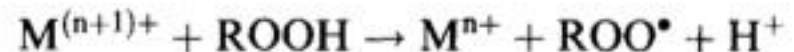
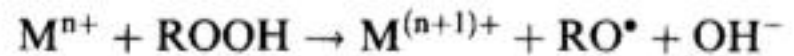
MICROWAVE CURED

15% w/w NH₂-PSF ($\overline{M}_n$ =16,200) into DGEBA / 4,4'-DDS

Anaerobic Adhesives



Free radical polymerizations
Free oxygen reacts with
radicals and makes them less
reactive.



Silicone Adhesives (reactive)

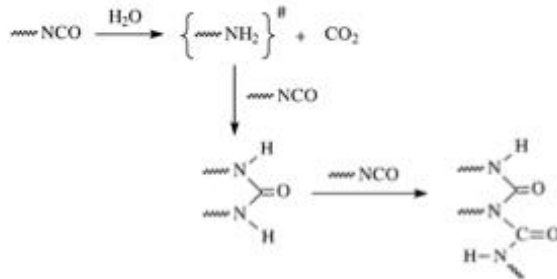
Primary Reaction



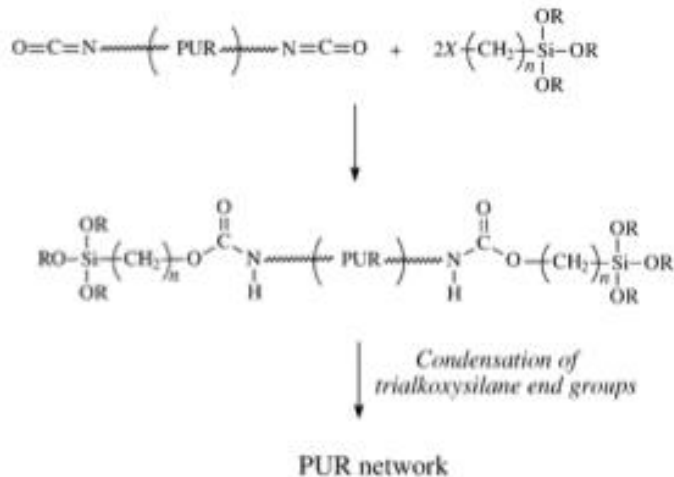
Secondary Reactions (Post-Cure)



Polyurethane Reactive (PUR) Moisture curing



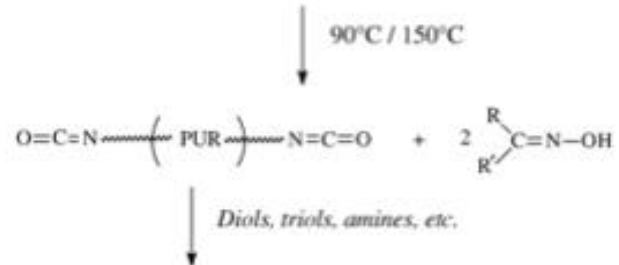
Reaction 3. Reactivity of isocyanates with water.



Reaction 5. Chain extension in hybrid α,ω -trialkoxysilane polyurethane (PUR).



Blocked polyurethane formulated with diols, triols, amines, etc.



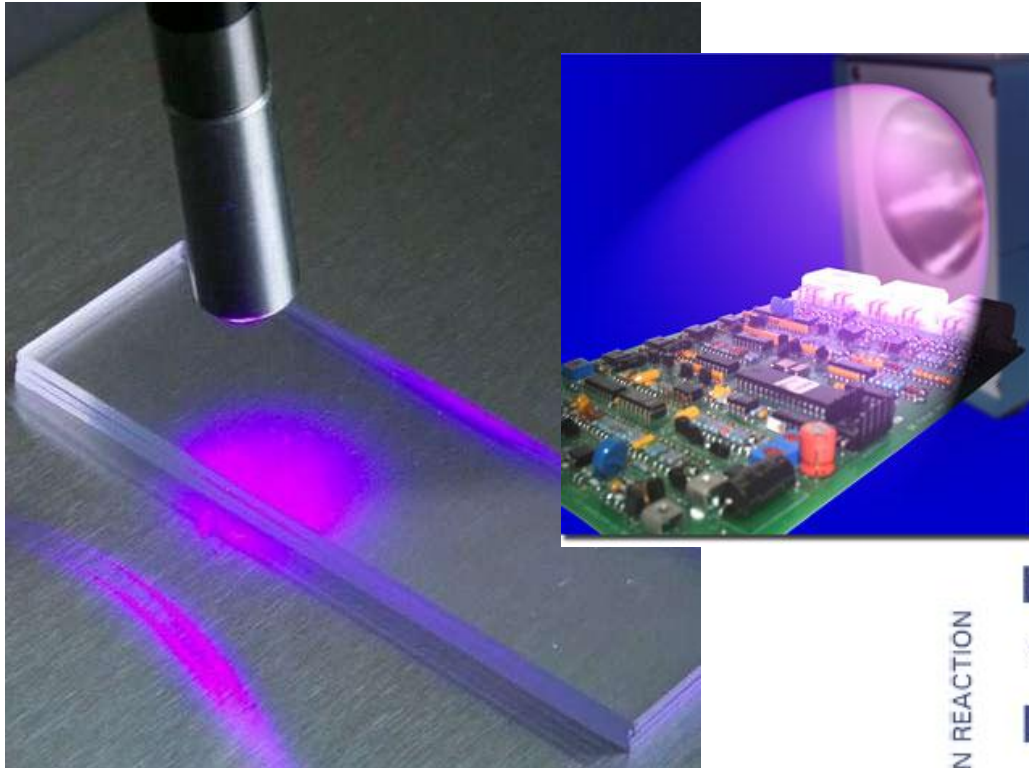
Chain extension and cross-linking

Reaction 4. Chain extension and cross-linking in blocked polyurethanes (PURs).

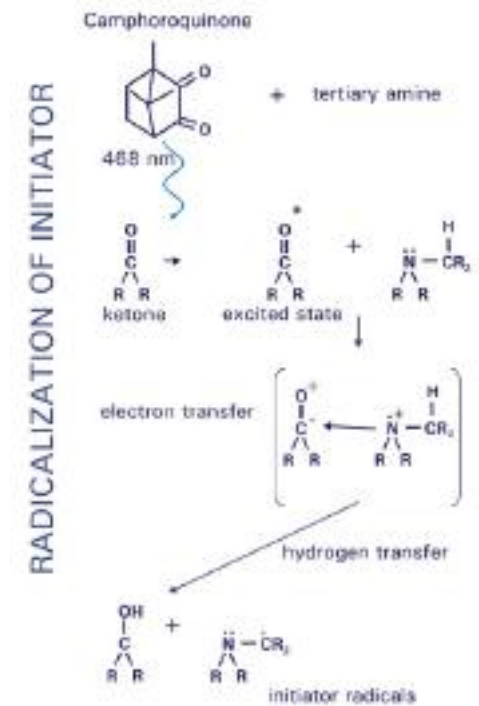


Polyurethane
glue –
Gorilla Glue

UV Cure

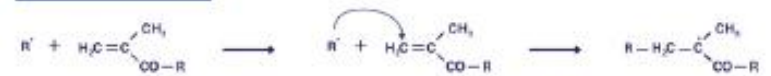


Crosslinker monomers react to give network very quickly

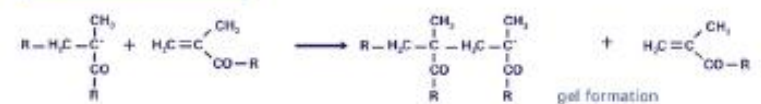


POLYMERIZATION REACTION

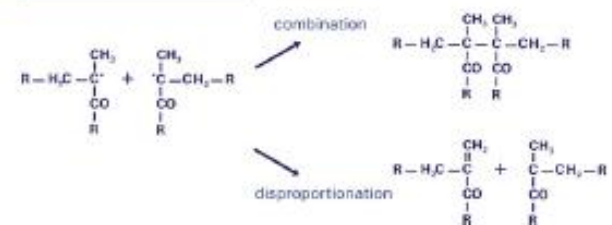
Initiation reaction



Propagation reaction

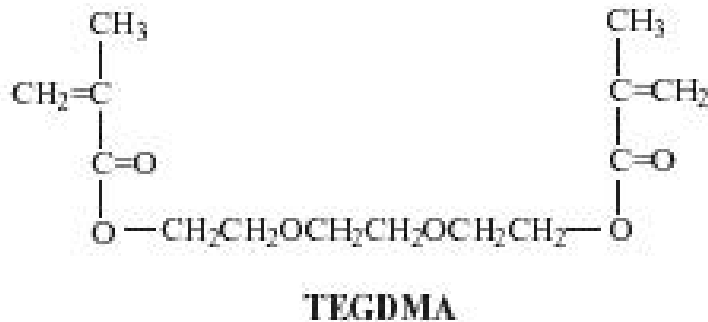
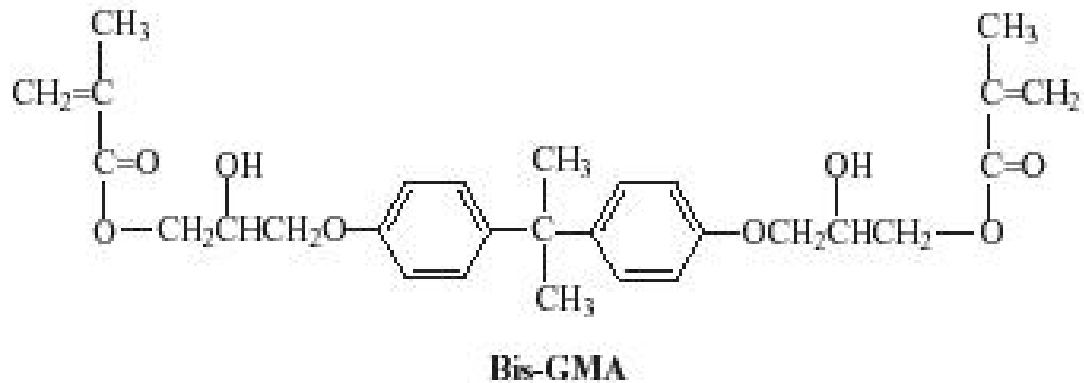


deactivation reaction



Most Common Dental Resin

bisphenol A glycol dimethacrylate



triethylene glycol dimethacrylate

Glass-ionomers



- Most commonly referred to as *glass ionomer cements* (GIC's).
- These are materials which consist of an aqueous polyacrylate and a fluoroaluminosilicate glass.
- These set by an acid-base reaction in the presence of water.
- These cements appear to adhere to tooth structure by formation of ionic bonds as a result of chelation of the carboxyl groups in the acid with the calcium and/or phosphate ions in the apatite of enamel and dentin.

COMPOSITION

POWDER = CALCIUM FLUOROALUMINOSILICATE GLASS

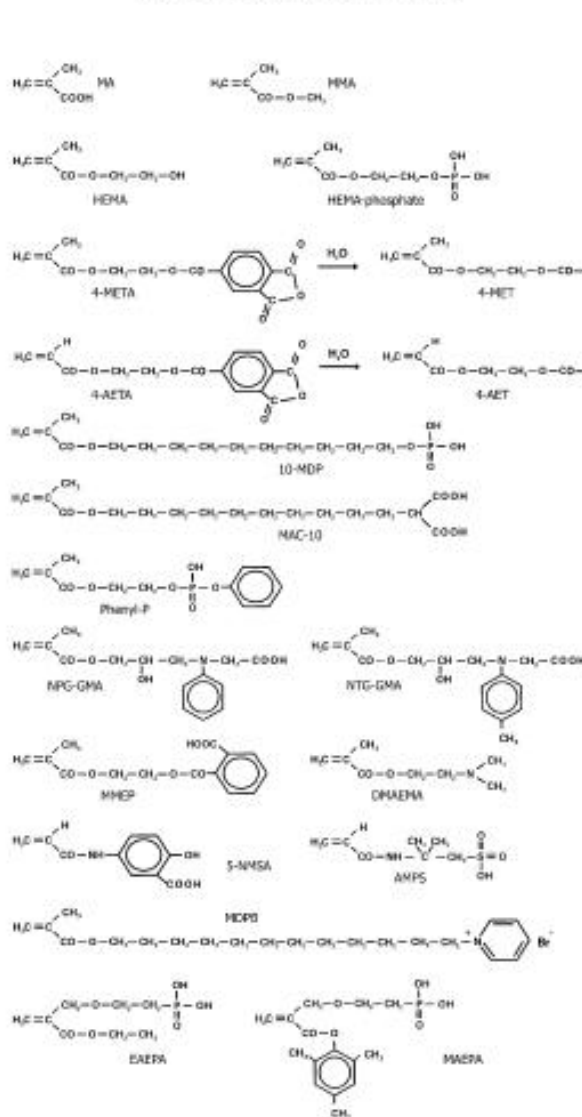
- Silicon 13.3% Phosphorus 2.5%
- Aluminum 13.3% Fluoride 22.7%
- Calcium 17.3% Oxygen 28.0%
- Sodium 1.6%

LIQUID = POLYACRYLIC ACID OR

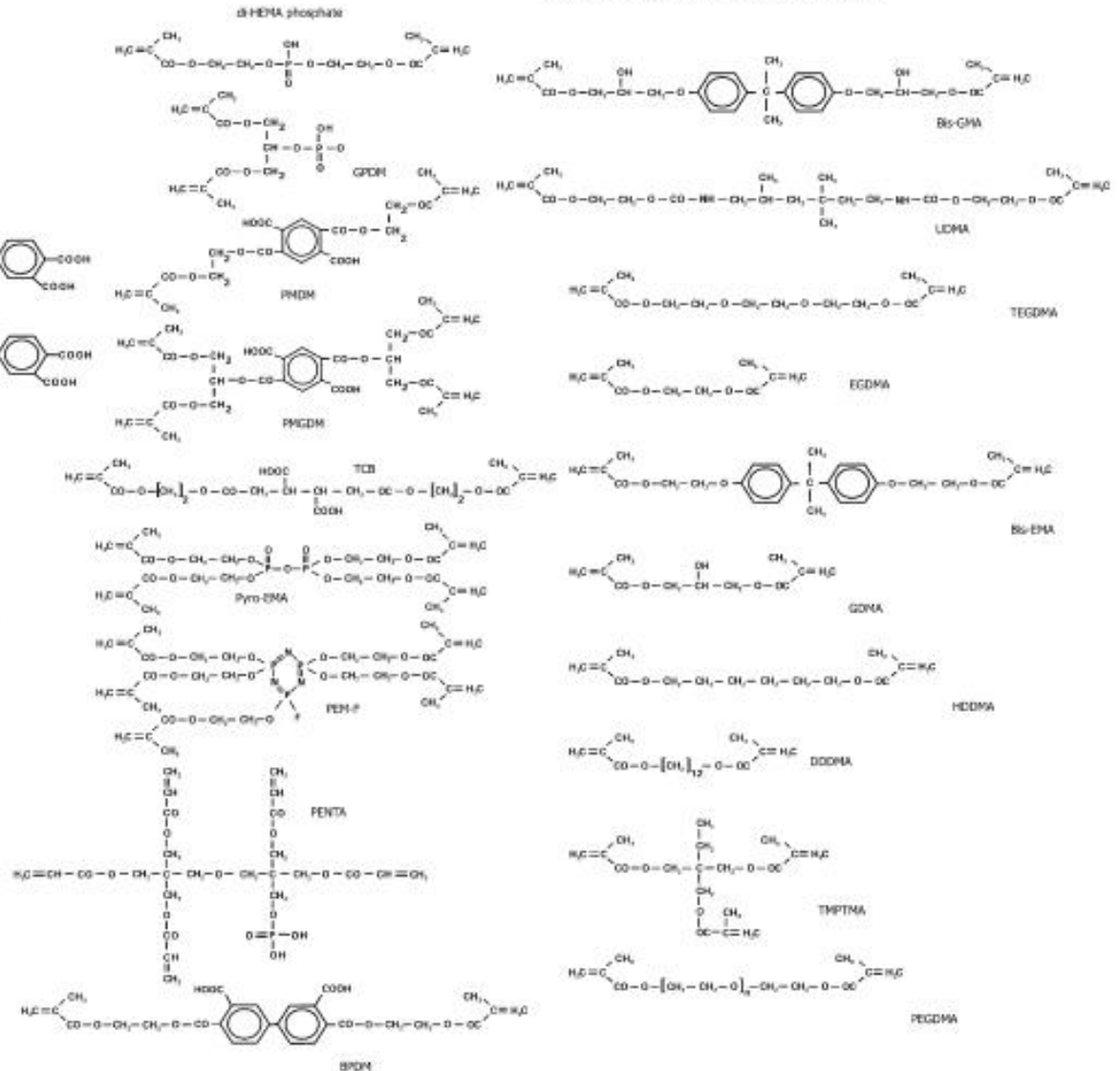
- COPOLYMER OF ACRYLIC ACID
- WATER

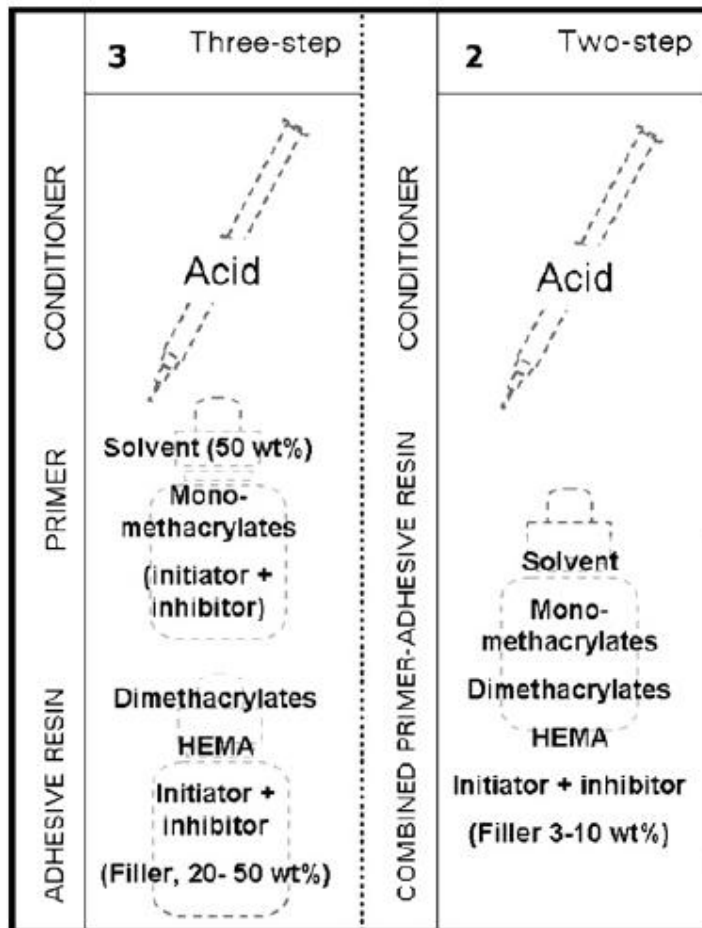
Dental composite resins

FUNCTIONAL MONOMERS

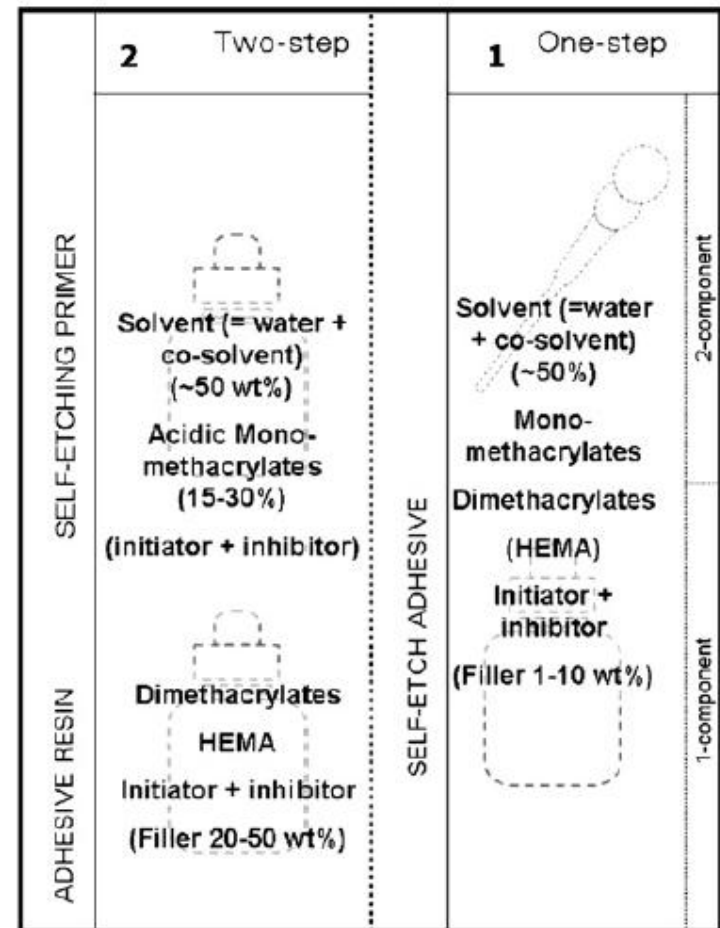


CROSS-LINKING MONOMERS





Etch & rinse adhesives

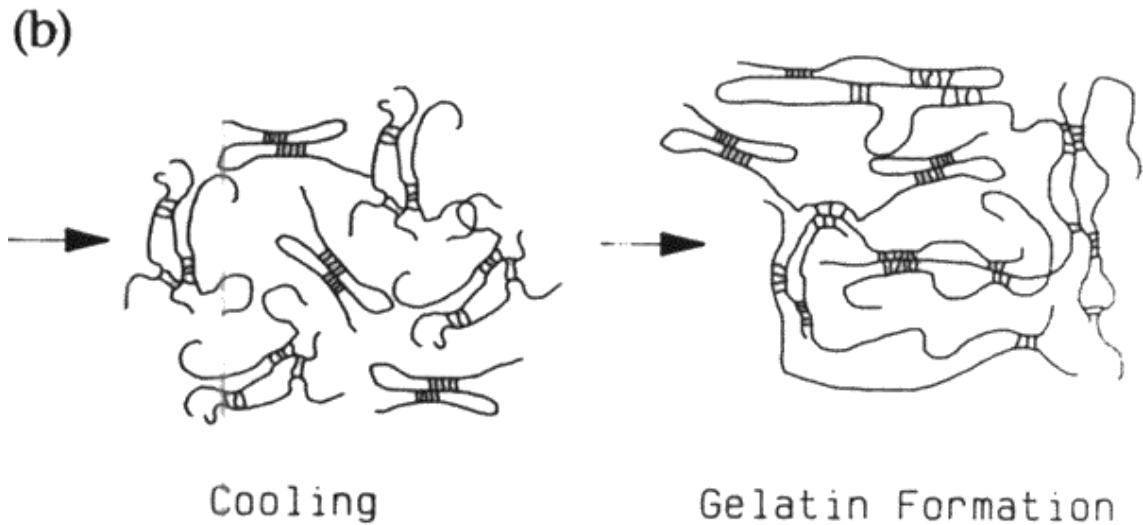
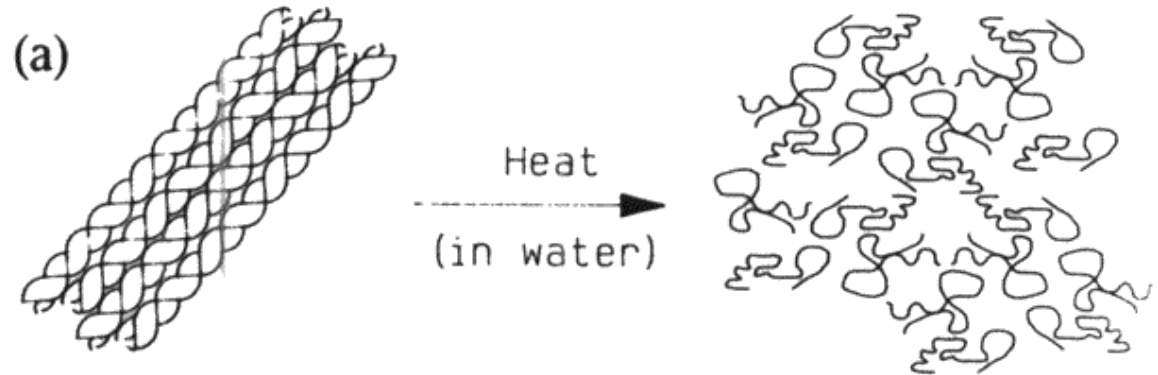


Self-etch adhesives

Natural Adhesives

- Spider web
- barnacles
- honey Dew
- asphalt
- egg white
- animal protein
- starch
- Shellac
- blood proteins
- dextrans
- animal and vegetable proteins
- Polyisoprene rubber

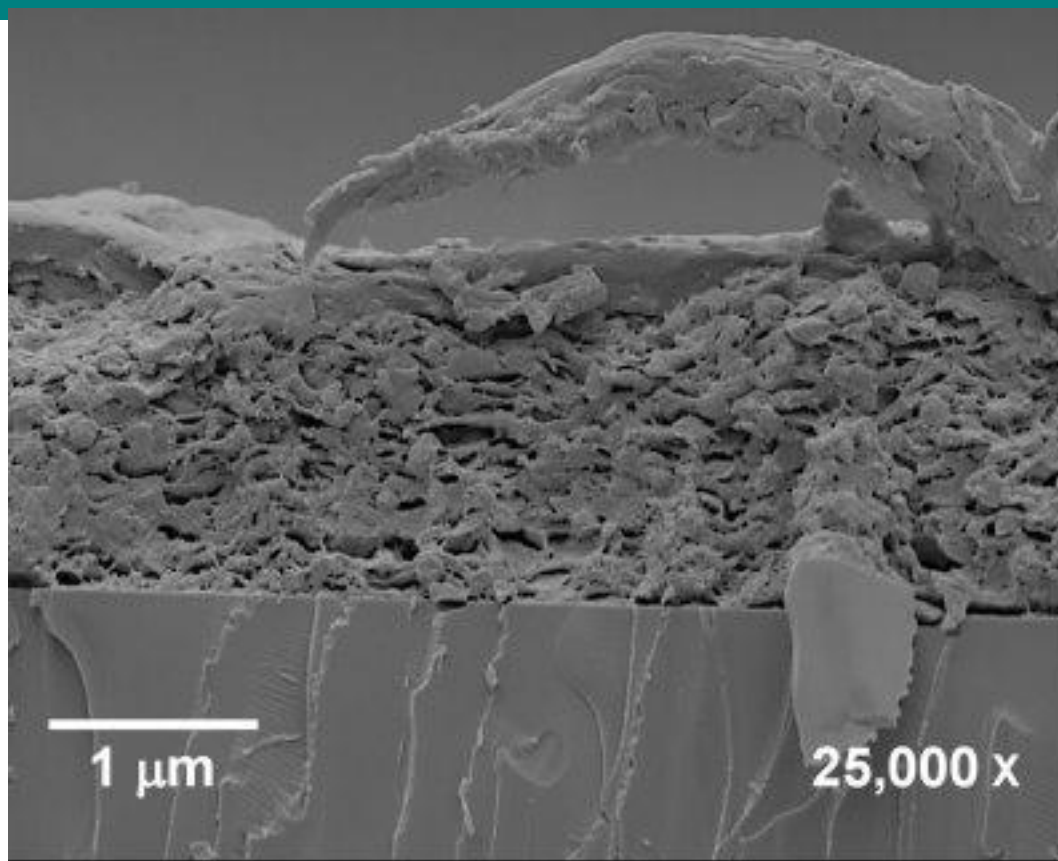
Animal Glue



Glue

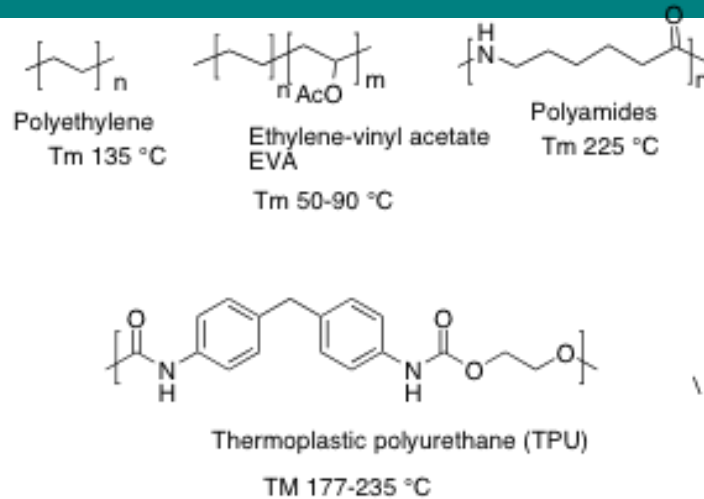
- Animal glue is an organic colloid derived from collagen
- Casein – manufactured from skim milk
- Other organic glues manufactured from soy beans, blood
- Gurry (old fish heads, skins, etc) – fish glue

Mussel glue



Tensile strength 0.2 MPa to 0.7 MPa following a 1 hr set time under water.

Hot melt adhesives



Construction

Manufacture of laminated wood panels; kitchen countertops.

Non-rigid Bonding

Bonding of woven and non-woven fabrics; manufacture of athletic shoes, books, and sporting goods.

Packaging

Manufacture of cartons, boxes and corrugated boards; bags, envelopes, disposable products (diapers, paper products); cigarettes; and labels, stamps.

Vehicles

Aircraft and aerospace structural assemblies; automotive, truck, boat, and bus assembly; mobile home manufacturing.

Hot Melt Adhesives



- Thermoplastics-below T_g
- Often polyethylene
- Melt polymer
- Must wet surfaces
- Cool polymer
- May involve crystallization-stronger structures
- Mechanical interlocking mechanisms



Hot Melt

Adhesive dispenses warm at 170°F (77°C) and can remain in the applicator at dispensing temperature for up to 40 hours.

Optional pre-heater keeps up to four adhesive cartridges ready to use.

Easy to load cartridge with plastic disposable nozzle allows easy, quick changeover.

Electrically-heated pneumatic applicator dispenses with the squeeze of a trigger.

Disposable nozzles for maintenance ease.

Impact-resistant plastic construction for lightweight handling ease and the hard realities of the shop.

How do adhesives work?

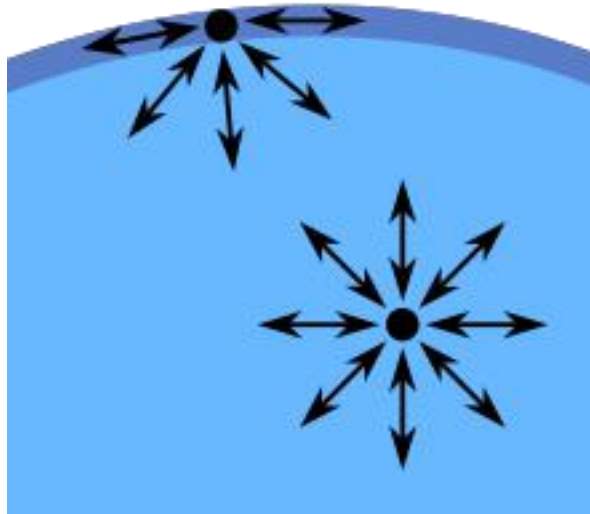


Two types of Adhesion

- Type I – Surface tension
 - Oil between precision ($\pm$ 50 millionths) machined steel blocks will glue them together
- Type II – mechanical interlocking
 - Most common
 - Rough surfaces



Surface Tension



LaPlace Equation

$$\gamma = \frac{\text{Energy (J)}}{\text{Area (m}^2\text{)}}$$

$$\Delta p = \gamma \left(\frac{1}{R_x} + \frac{1}{R_y} \right)$$

Δp is the pressure difference.

γ is surface tension.

R_x and R_y are radii of curvature in each of the axes that are parallel to the surface.

Surface tension is the force per unit length that must be applied parallel to the surface so as to counterbalance the net inward tension of the surface (units dynes/cm or N/m; 1000 dyne/cm = 1 N/m) Alternatively, it is the energy required too increase the surface area of a liquid (or phase).

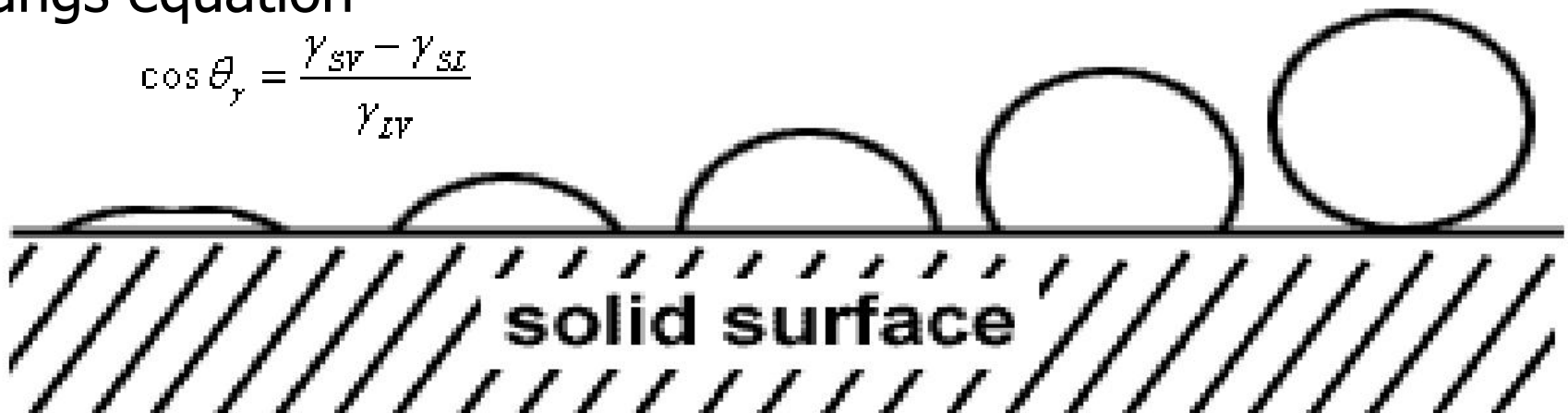
**complete
wetting**

incomplete wetting

**no
wetting**

Youngs equation

$$\cos \theta_y = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}}$$



$\theta = 0^\circ$

$\theta < 90^\circ$

$\theta = 90^\circ$

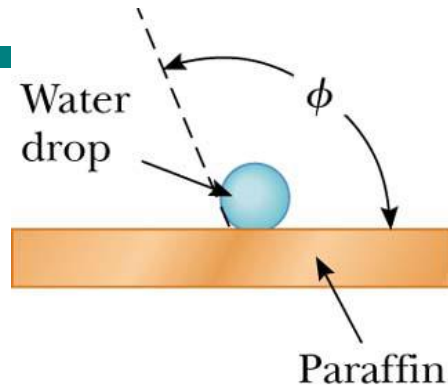
$\theta > 90^\circ$

$\theta = 180^\circ$

γ_{sv} = solid vapor surface tension γ_{sl} = solid liquid surface tension

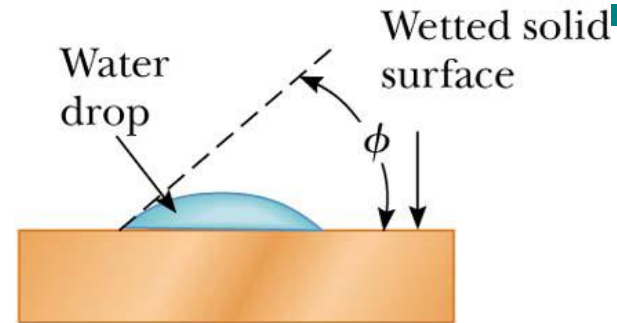
γ_{lv} = liquid vapor surface tension

More on surfaces



(a)

© 2003 Thomson - Brooks/Cole

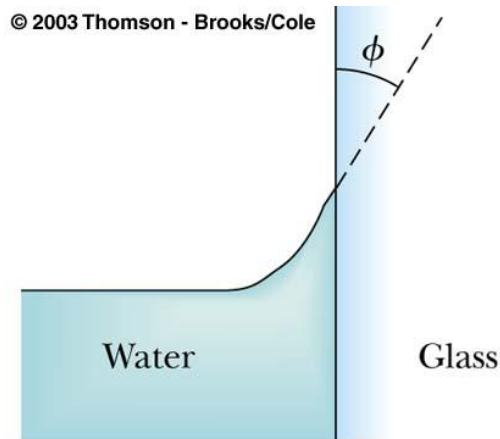


(b)

Liquid wets the surface if $0 < q_c < 90^\circ$

If cohesive forces are stronger than the adhesive ones like molecules in the drop try to stay together to reduce the total energy of the system; if adhesive forces are stronger the drop will spread out to reduce the total energy of the system. The spreading will stop when the surface tension becomes too strong.

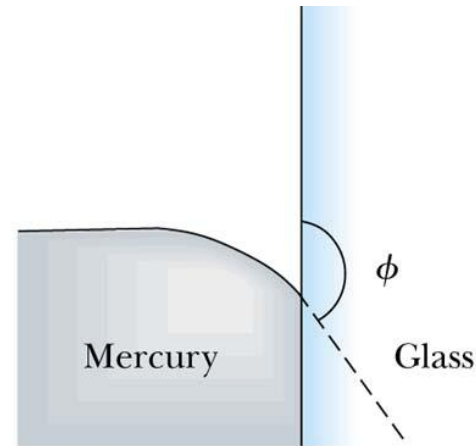
Same thing



(a)

Adhesive > Cohesive

The water wants to cover as much of the glass as its surface tension allows



(b)

Adhesive < Cohesive

The mercury wants to cover as little of the glass as its surface tension allows

$$\gamma_{\text{ADHESIVE}} \leq \gamma_{\text{SUBSTRATE}}$$

Liquid	Surface Tension (dyne/cm)
Water	73
Glycerol	63
Ethylene Glycol	48
Nitrobenzene	44
Epoxy Resin	43
Petroleum Lubricating Oil	29
Benzene	29
Silicone Oil	21
n-Hexane	18

Solid	Critical Surface Tension (dyne/cm)
Copper	~ 1000
Aluminum	~ 500
Glass	290
Acetal	47
Polyamide	46
Polycarbonate	46
Polyethylene Terephthalate	43
Cured Epoxy Resin	43
Polyimide	40
Polyvinyl Chloride	39
Polystyrene	33
Polyethylene	31
Silicone	21
Polytetrafluoroethylene	18

Surface Tension of Polymers

Table 6.1 Surface tension values for some plastics	
Material	Surface tension (mN/m)
PTFE	18
Acetal	22
Polypropylene	29
Polyethylene	31
Polystyrene	35–37
Polymethylmethacrylate (acrylic)	39
PVC	39
Polyethylene terephthalate	41–47
Polycarbonate	46
Nylon 6	46

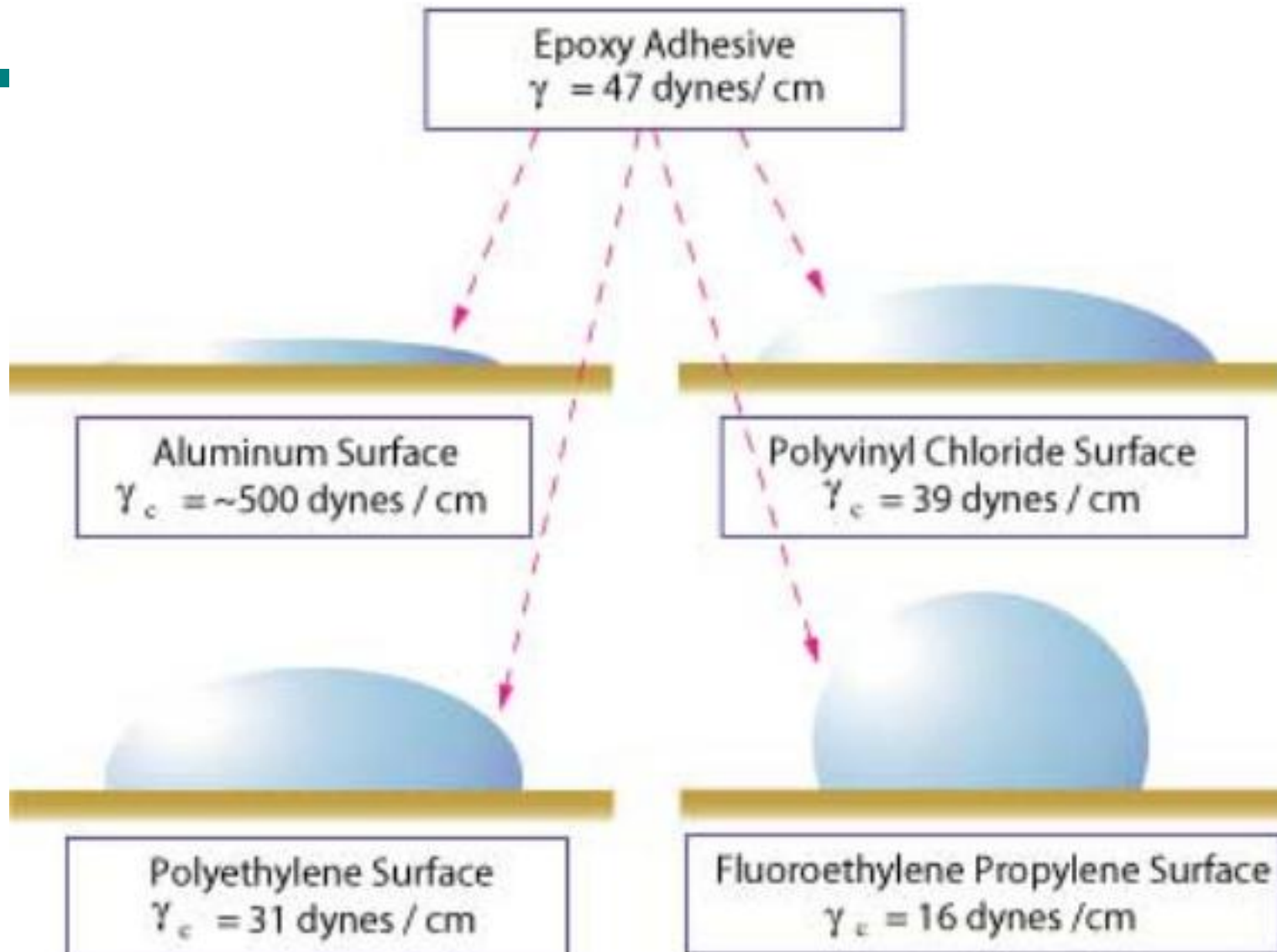
Epoxy Adhesive
 $\gamma = 47$ dynes/cm

Aluminum Surface
 $\gamma_c = \sim 500$ dynes/cm

Polyvinyl Chloride Surface
 $\gamma_c = 39$ dynes/cm

Polyethylene Surface
 $\gamma_c = 31$ dynes/cm

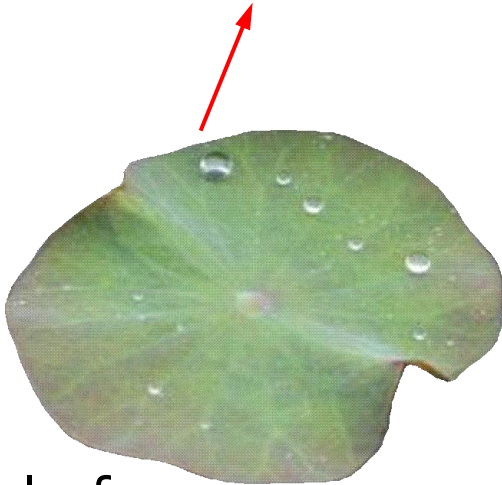
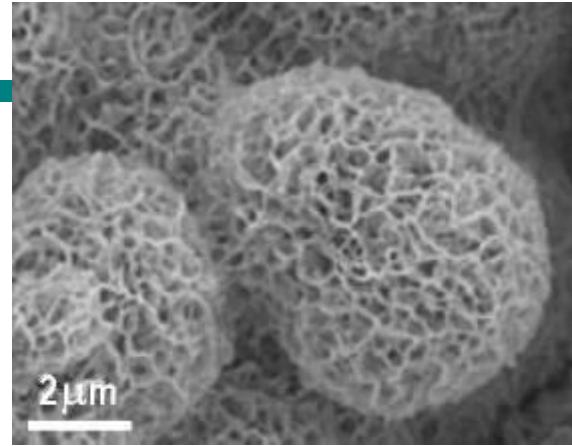
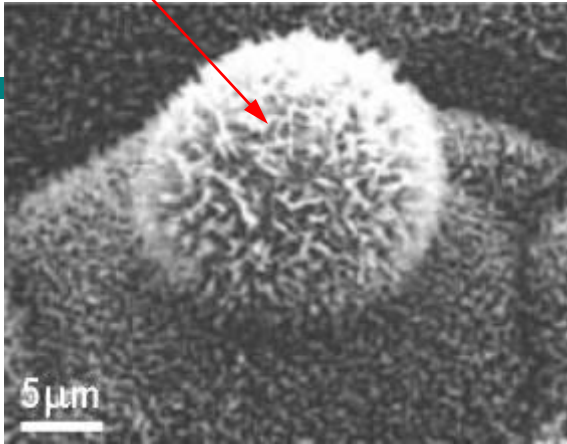
Fluoroethylene Propylene Surface
 $\gamma_c = 16$ dynes/cm



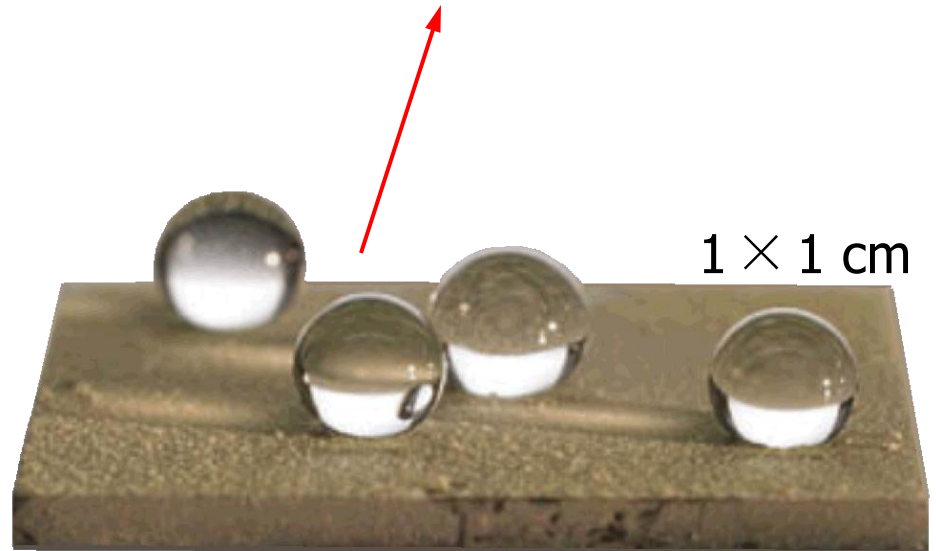
The Lotus-Effect



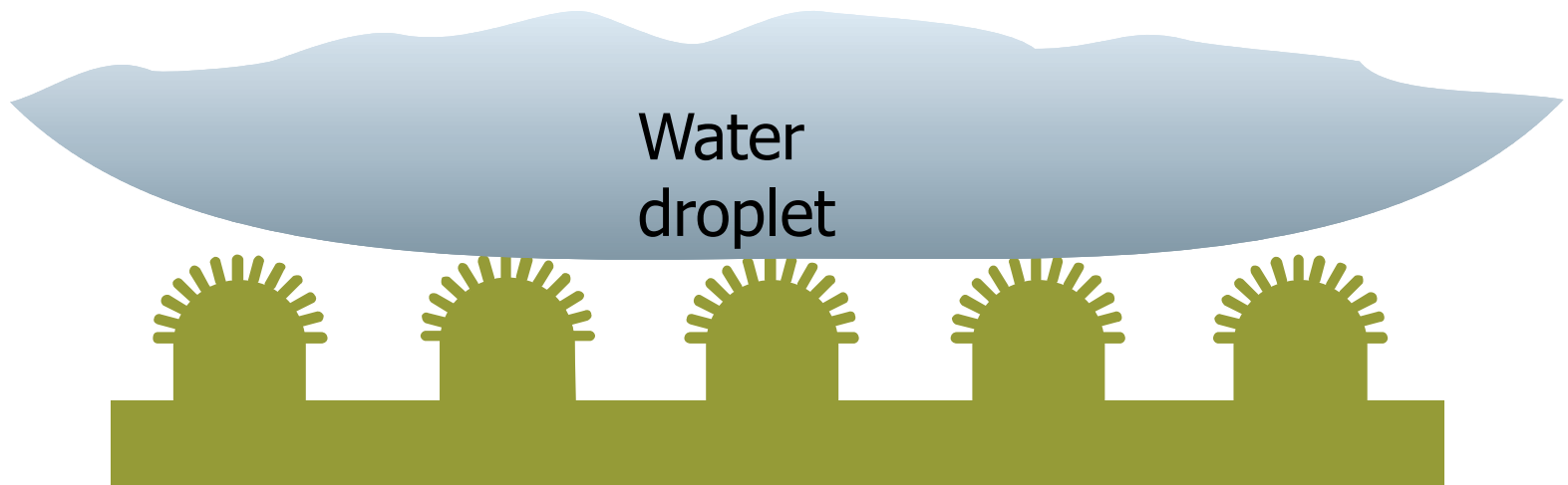
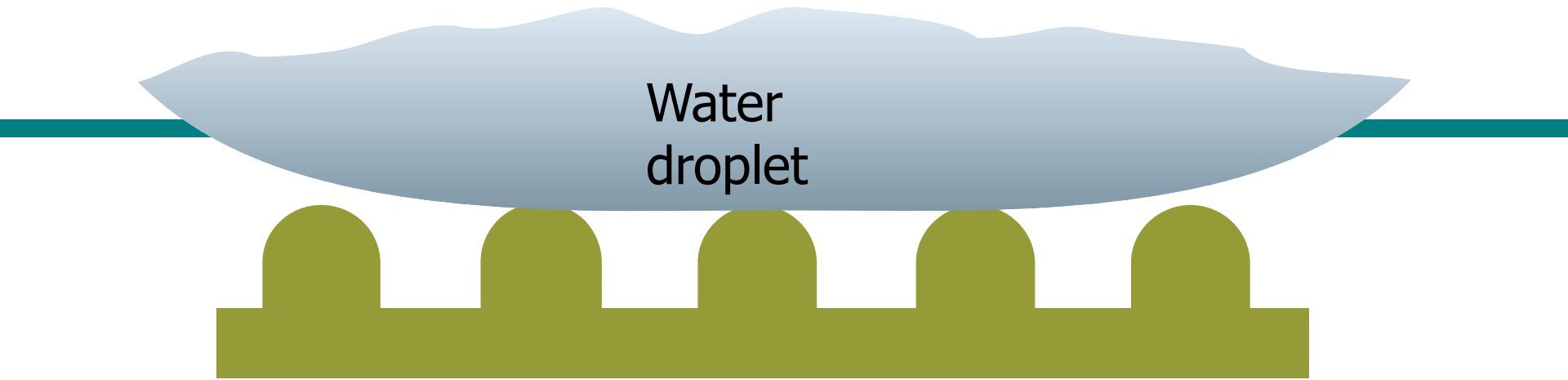
Secondary structure

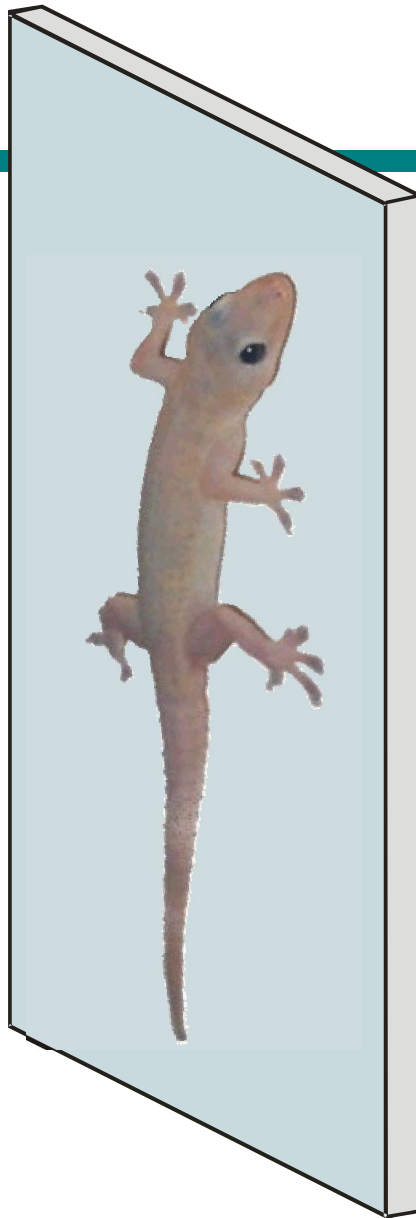


Lotus leaf



engineering imitation





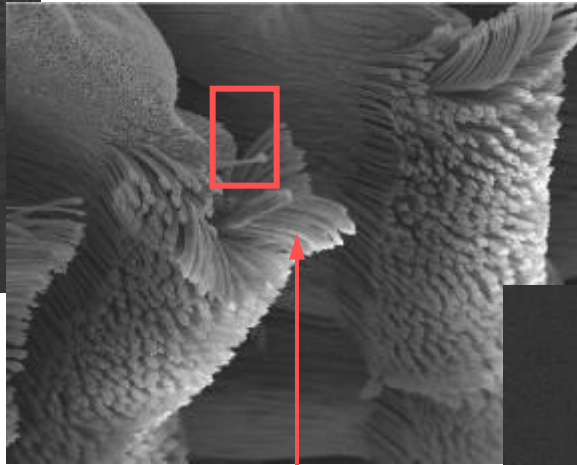
500 000 000 nanohairs



Geckos get a grip using
Van-der-Waals-forces

Gecko sticking at the wall
In fact, a single gecko foot can
suspend a 40 lb weight

Nanostructure of the Gecko toe



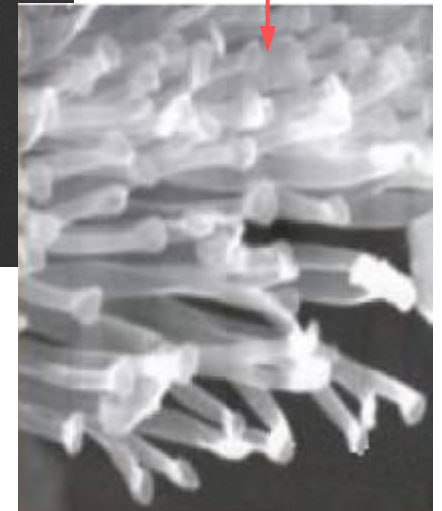
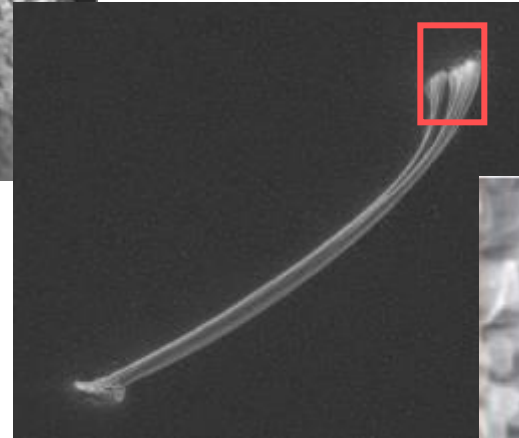
The Gecko toe has 500000 microhairs (setae)

One setae can theoretically lift 200 mg (200 mN)

10 N/cm^2

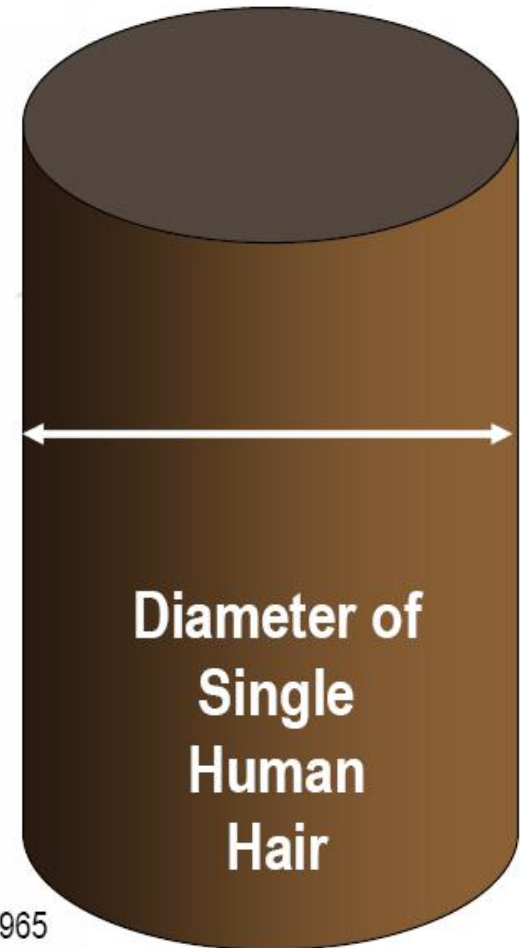
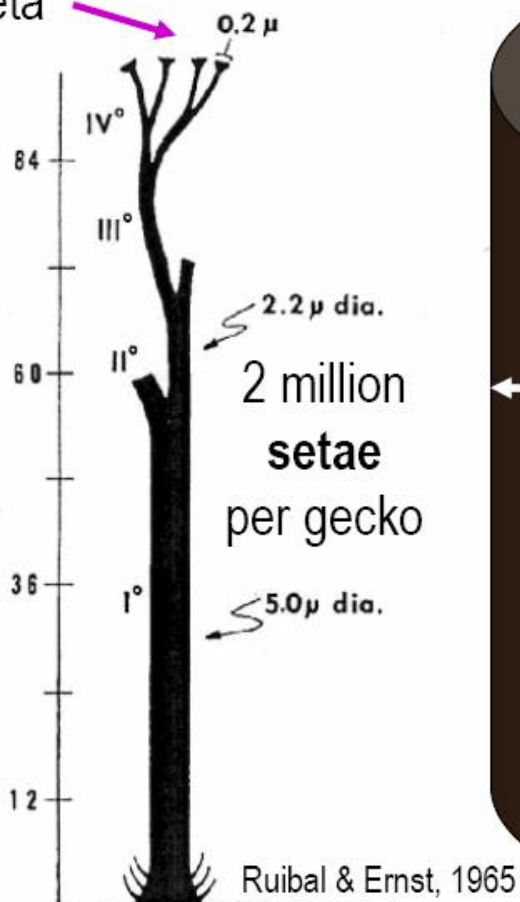
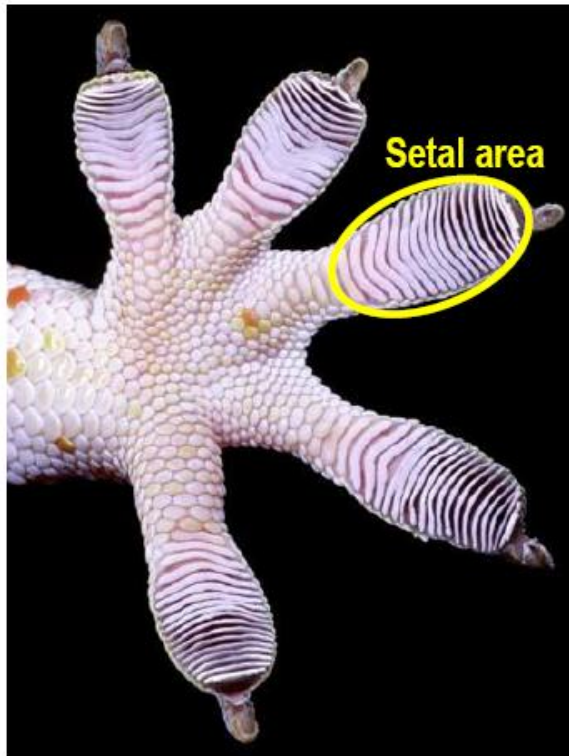
50-300 nN/pillar

The seta has 1000 nanohairs

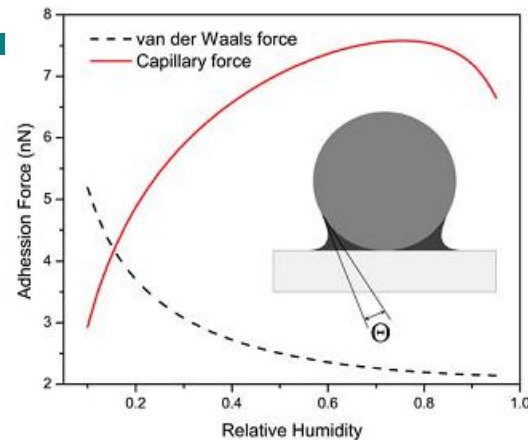
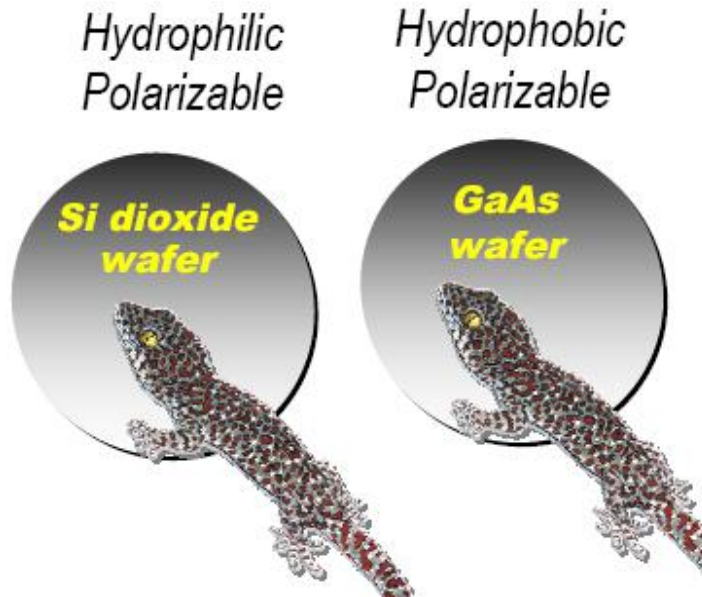


Billions of nanohairs per Gecko

100-1000 **spatulae** per seta
($10^8 - 10^9$ per gecko)



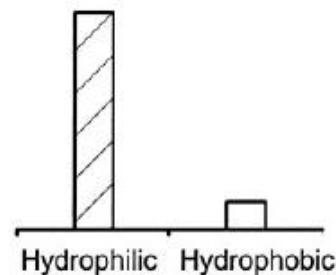
Capillary Forces vs Van der Waals



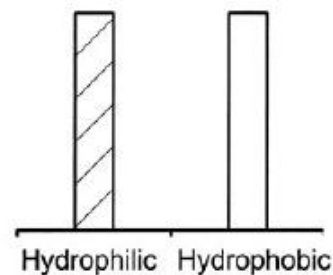
Capillary Action (Thin Film Adsorption) requires hydrophilic (not hydrophobic) substrate

Full PNAS (2002)

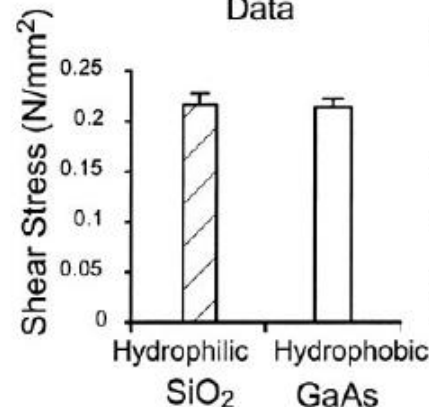
A Capillary Adhesion Prediction



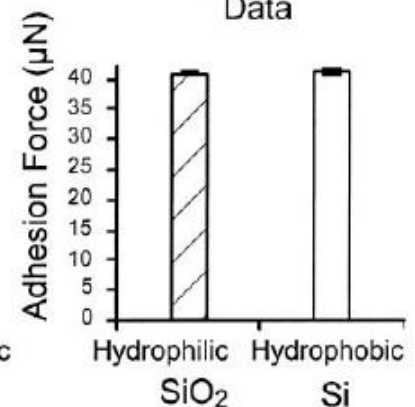
B Van der Waals Adhesion Prediction



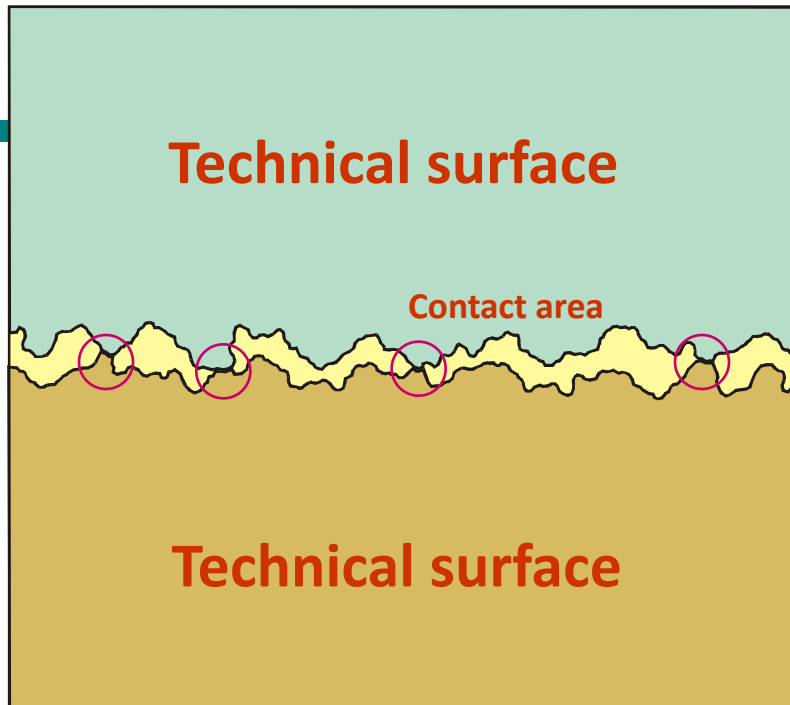
C Semiconductor Wafer Data



D MEMS Cantilever Data



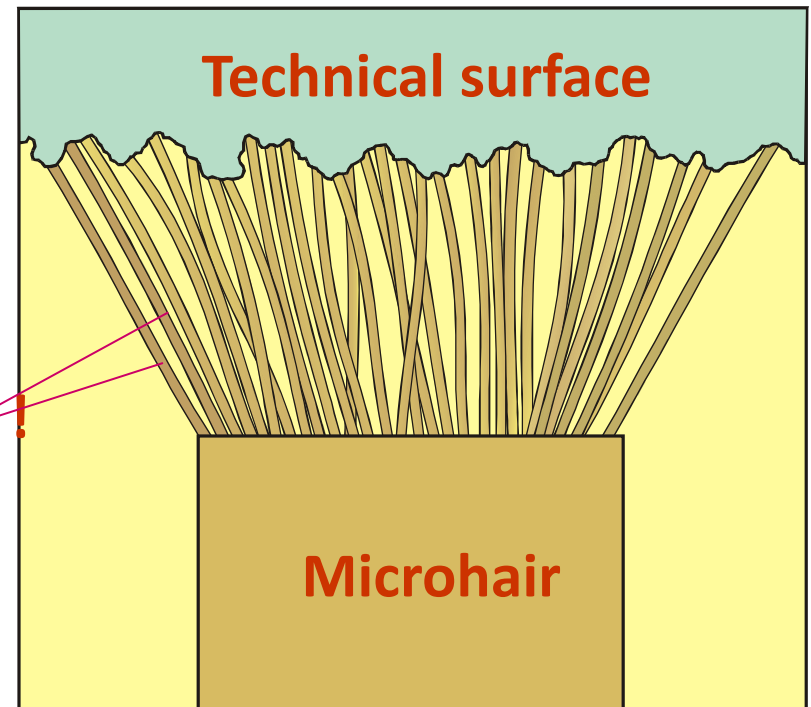
The Gecko effect



Small contact area
small adhesion force

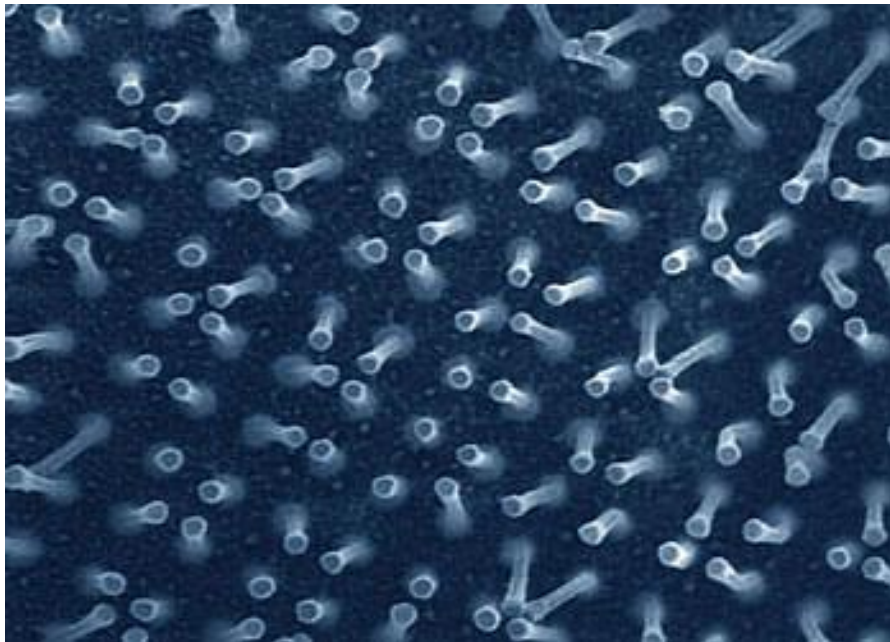
Nanohairs!

Adhesion effect through
Van-der-Waals-forces



Large contact area
large adhesion force

Gecko-Tape



29 N/cm²



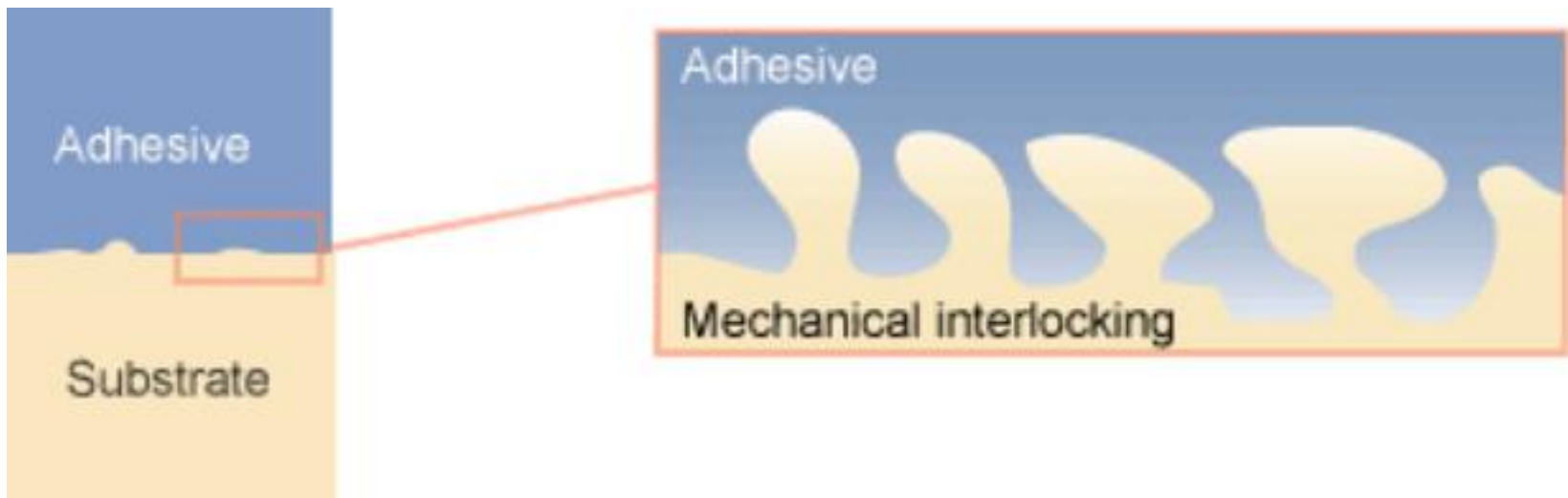
Qu and L. Dai, Adv. Mater. **19**, 3844 (2007).

Adhesive Viscosity

- Must be low enough to allow flow into joint and into rough surfaces and pores
- Must increase as much as possible to maximize bond strength



Mechanical Interlocking (Type II)



Increase surface area at interface
to improve mechanical bonding

Surface Treatment

- No treatment (low cost poor reproducibility);
- Solvent wiping;
- Vapor degreasing;
- Mechanical abrasion;
- Plasma treatment;
- Etching;
- Chemical deposition-primers, organosilanes

Surface Treatments for Polymeric Materials

Polymeric sheets/films, composites, polymeric coatings

<u>Adherend</u>	<u>Clean</u>	<u>Treatment</u>	<u>Adhesive</u>
polyamide	ketone	abrasion, primer	phenol- (Nylon) formaldehyd, epoxy, CN acryl.
polycarbonate	alcohol	abrasion	epoxy, urethane, CN acryl.
polyimide	ketone	abrasion, NaOH, plasma	epoxy
polystyrene	alcohol	abrasion	urethane, epoxy, CN-acryl.
polysulfone	alcohol	abrasion	urethane,epoxy

Surface Treatments for Polymeric Materials

Polymeric sheets/films, composites, polymeric coatings

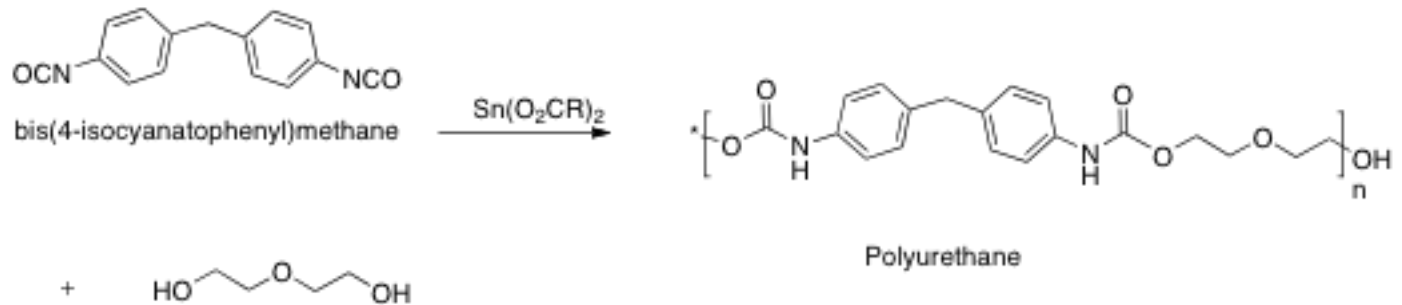
Adherend	Clean	Treatment	Adhesive
PVC, PVF	ketone	abrasion	epoxy, urethane, CN-acryl., nitrile-phenolic
F-carbons	Cl-alcohol	Na-NH ₃ ; Na naphthalene/THF, plasma	epoxy, urethane
polyethylene, polypropylene	ketone	flame, plasma, elec. discharge, acid dichromate	epoxy, nitrile-phenol
PET	ketone	NaOH soln., plasma	epoxy, urethane, NCO-p-ester

Adhesives for metals

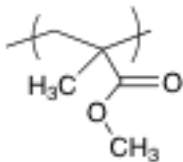
Epoxy



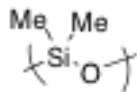
Polyurethane



Acrylic

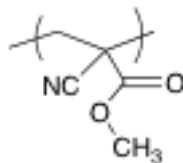


Silicone

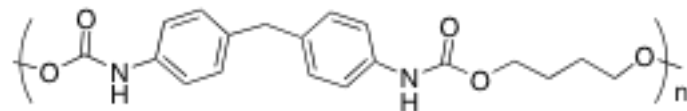


Adhesives for plastics

Cyanoacrylate



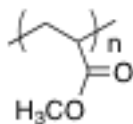
Polyurethane



Epoxy

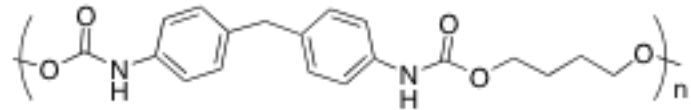


Acrylic

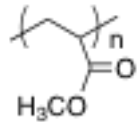


Adhesives for ceramics

Polyurethane



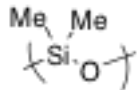
Acrylic



Epoxy

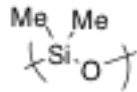


Silicone



Adhesives for rubber

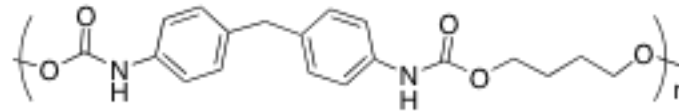
Silicone



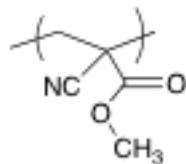
Epoxy



Polyurethane



Cyanoacrylate



Adhesives for wood

Formaldehydes (Phenol Formaldehyde

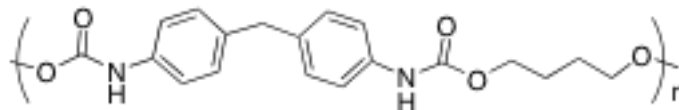
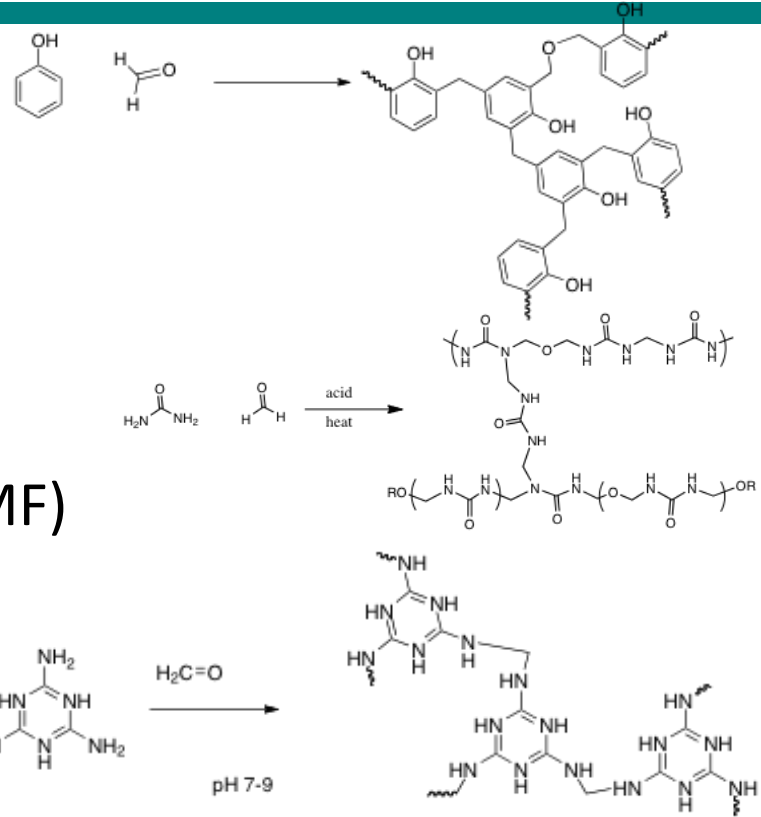
Thermoset Urea Formaldehyde (UF)

Thermoset Melamine Formaldehyde (MF)

Polyvinyl acetate (PVA, White Glue)

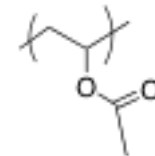
Aliphatic resin (Yellow Glue)

Polyurethane



Adhesives for paper

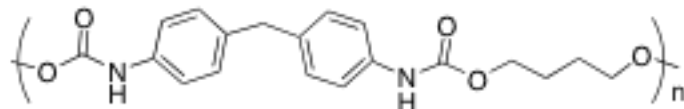
Polyvinyl acetate (PVA, White Glue)



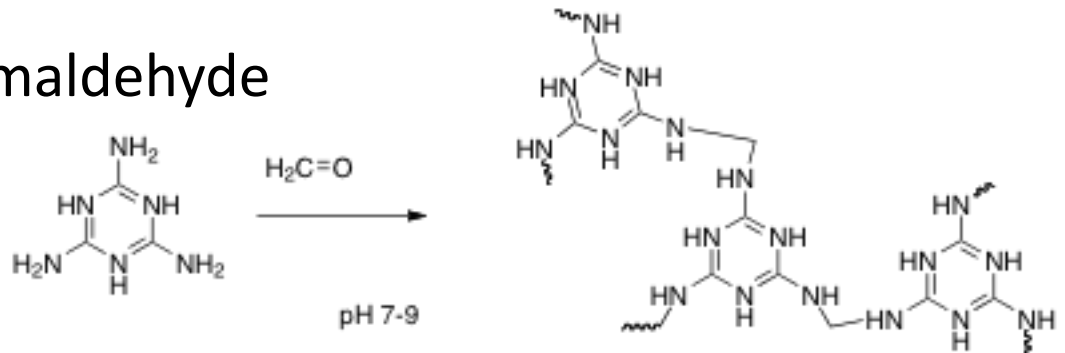
Epoxy



Polyurethane

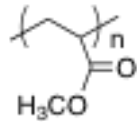


Thermoset Melamine Formaldehyde
(MF)



Adhesives for glass

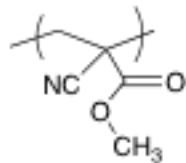
Acrylic



Epoxy



Cyanoacrylate



Durability of Adhesives

Table 14.1 Durability of wood joints bonded with various adhesives

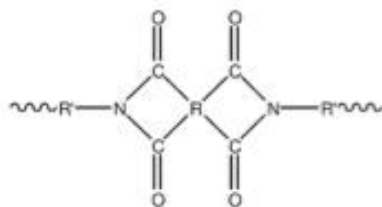
Type of adhesive	Full exterior conditions	Semi-exterior and damp interior conditions	Dry interior conditions	Class (BS 1204)
PF, RF and combinations	25 years expected	Indefinitely	Indefinitely	WBP
MF, MF:UF, Fortified UF	5–10 years	10–20 years estimated		BR
UF	2–5 years	5–10 years		MR
Casein	1–2 years	2–5 years		—
PVA	Comparable with casein ¹			—
Animal	Fail in a few months	1 year		—

¹ FPRL Bulletin no. 20

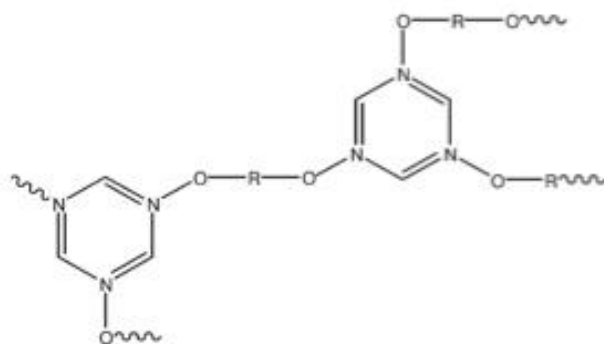
High-temperature Adhesives



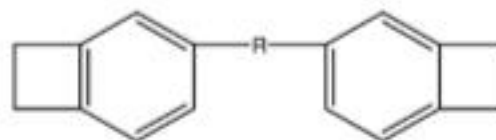
(3)



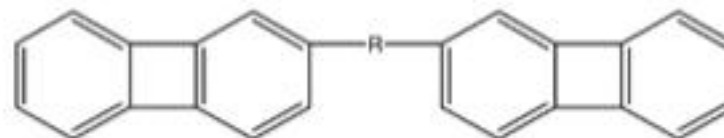
(4)



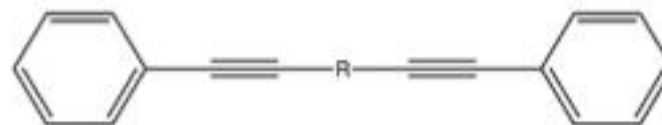
(5)



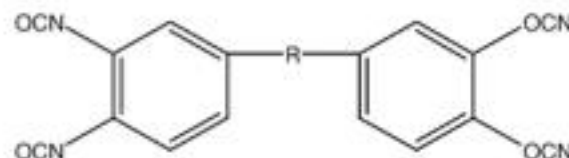
(6)



(7)



(8)



(9)

Chart 2. Examples of high-temperature adhesives: (3) bismaleimide, (4) polyimide, and (5) cyanate ester.

HOMEWORK

- Find 2 types of glues through Internet, briefly state their principles(formulation) and applications